PDE - Ordinary differential equations

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1 Introduction

Ordinary differential equation (ODE) : equation that relates a function with its derivatives (with respect to **one** variable). The unknown is a function.
They are useful for modeling, where one has to relate quantities and their variation.

Closed form : explicit formula which relates the solution y to the variable t
Warning : in general, an ODE *cannot* be solved by finding a closed form
Example : $y'' = -\sin(y)$

When considering an ODE, we are interested in finding :

1. if a solution y exists around the initial condition y_0
2. if a solution exists for a "large" set of $t \to$ are there any patterns such as periodicity?
3. if a small change in y_0 results in a small change of the solution
4. an estimate of y (computed numerically)

2 Initial Value Problem

2.1 The use of ODEs in modeling

In many fields, the modeling of problem leads the ODEs (simple pendulum, preys and predators populations, chemical reactions...).

Nondimensionalization/scaling : when considering a physical problem, it is useful to replace dimensional variables with adimensional counterparts. This process is known as nondimensionalization or scaling.
In order to nondimensionalize a variable X, we define X as $X = X^*\mathcal{X}$, with X^* a dimensionless variable and $\mathcal{X}$ a coefficient containing the dimension.

2.2 The Picard-Lindelhöf-Cauchy-Lipschitz Theorem

Cauchy Problem/Initial Value Problem (IVP) : problem consisting of an ODE and a specified value of the unknown function at a given point, called the **initial condition**

When modeling, this description is quite useful : the initial condition will describe the state of the system at a given time while the ODE will allow us to determine the evolution of the system.

Order of an ODE : highest derivative of the unknown function that appears in the ODE.

In an **ODE**, the derivative is with respect to **one variable** (usually the time t in our problems).
When there are derivative with respect to **multiple variables**, we speak of **partial differential equation (PDE)**.

Usual notations :
The unknown function is usually denoted y or u.
The variable is usually denoted x or t.

For the following, let's consider :

- $t_0 \in \mathbb{R}$
- $I \subseteq \mathbb{R}$ open interval containing t_0
- $\mathcal{U}$ open set of $\mathbb{R}^d$, $\quad d \in \mathbb{N}^*$
- $f : I \times \mathcal{U} \to \mathbb{R}^d$
- $y_0 \in \mathcal{U}$
- the problem $\mathcal{S} : \begin{cases} y'(t) = f(t, y(t)), & t \in I \\ y(t_0) = y_0 \end{cases}$

Note : this situation deals with ODE of order higher than one using the matrix notation
Example : $\theta'' + k\theta = 0 \iff \begin{bmatrix} \theta' \\ \theta'' \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ k & 0 \end{bmatrix} \cdot \begin{bmatrix} \theta \\ \theta' \end{bmatrix}$

Slope field : the slope field associated to the ODE is $\begin{cases} I \times \mathcal{U} \to \mathbb{R}^2 \\ (t, x) \mapsto \begin{bmatrix} 1 \\ f(t, x) \end{bmatrix} \end{cases}$
We often normalize the slope field for representation purposes.

Integral curve/orbit : an integral curve of the slope field is the curve of a function y solution to the IVP.
We can represent several orbits on the same graph (for various initial conditions).

Note : the vectors of the slope field are tangents to the orbits.

Proving the existence of solutions for an IVP :
Local Picard-Lindelhöf/Cauchy-Lipschitz Theorem :
Consider the IVP $\mathcal{S} : \begin{cases} y'(t) = f(t, y(t)), & t \in I \\ y(t_0) = y_0 \end{cases}$
Suppose :

1. $\forall x, t \mapsto f(t, x)$ is continuous
2. $\forall t, x \mapsto f(t, x)$ is $\mathcal{C}^1$

Then :

1. $\exists \varepsilon > 0, \mathcal{S}$ has a unique solution y on $[t_0 - \varepsilon, t_0 + \varepsilon]$

Theorem of alternatives : f uniformly bounded over $\mathcal{I} \times \mathbb{R}^d$ (i.e. $||f||_{\infty, \mathcal{I} \times \mathbb{R}^d}$ exists) $\implies$ the solution is global.
Note : this theorem is not mentioned in the videos, and appears only in the lecture notes

Note : **Integro-diffential form** (second form) :
Let $y \in \mathcal{C}^1([0,T])$.
$$\begin{cases} y'(t) = f(t, y(t)) \\ y(t_0) = y_0 \end{cases} \iff \forall t \in [0,T], \quad y(t) = y_0 + \int_{t_0}^{t} f(s, y(s)) ds$$

Gronwall's inequality :
Let $T > 0$.
Suppose :

1. $\phi, \psi : [0,T] \to \mathbb{R}_+$ are continuous
2. $\exists a \geq 0, \forall t \in [0,T], \phi(t) \leq a + \int_0^t \phi(s)\psi(s) ds$

Then :

1. $\forall t \in [0,T], \quad \phi(t) \leq a\, exp(\int_0^t \psi(s) ds)$

Global Picard-Lindelhöf/Cauchy-Lipschitz Theorem :
Consider the IVP $\mathcal{S}$: $\begin{cases} y'(t) = f(t, y(t)), \quad t \in I \\ y(t_0) = y_0 \end{cases}$
Suppose :

1. $\forall x, t \mapsto f(t,x)$ is continuous
2. $\forall t, x \mapsto f(t,x)$ is **globally Lipschitz continuous** with a Lipschitz constant independent of t (i.e. $\exists L > 0, \forall t \in I, \forall x_1, x_2 \in \mathcal{U}, ||f(t,x_1) - f(t,x_2)|| \leq L||x_1 - x_2||$)

Then :

1. The IVP has a unique solution y on I

Well-posedness : an ODE is well-posed in the sense of Hadamard if :

1. a solution exists
2. the solution is unique
3. the solution's behavior changes continuously with respect to the initial conditions

An ODE that is not well-posed is said to be **ill-posed**.

Regularity :
Let $k \geq 1, f \in \mathcal{C}^k(I \times U)$.
The solution to $\begin{cases} y'(t) = f(t, y(t)) \\ y(t_0) = y_0 \end{cases}$ is in $\mathcal{C}^{k+1}(J)$

Geometrical interpretation of the Picard-Lindelhöf theorem :
On the slope field, the orbits of two different initial condition can never intersect (this would violate the uniqueness).

Flow :
Let $f \in \mathcal{C}^2(I \times \mathcal{U})$.
$\forall (t_0, y_0) \in I \times \mathcal{U}$, $\exists J \times \mathcal{V}$ neighborhood of (t_0, y_0) and $\phi_{t_0} \in \mathcal{C}^1(J \times \mathcal{V}, \mathcal{U})$, $\begin{cases} \forall (t,x) \in J \times \mathcal{V},\ \phi'_{t_0}(t,x) = f(t, \phi_{t_0}(t,x)) \\ \forall x \in \mathcal{V},\ \phi_{t_0}(t_0, x) = x \end{cases}$
ϕ_{t_0} is called the (local) flow of the ODE at t_0

3 Linear ODEs

3.1 First order $(p = 1)$ and scalar $(d = 1)$

Linear ODE : an ODE is linear is it is linear in the unknown function and its derivatives.

Linearization : process of turning a non-linear ODE into a linear one by making an approximation.
Example : the simple pendulum equation can be linearized when θ is small...

Finding the closed form :
Let a, b be continuous functions on $\mathbb{R}$.
$\begin{cases} y'(t) = a(t)y(t) + b(t) \\ y(0) = y_0 \in \mathbb{R} \end{cases}$ has one solution on $\mathbb{R}$, given by $\forall t \in \mathbb{R}, \quad y(t) = e^{A(t)}(e^{-A_0}y_0 + B(t))$

A is the **anti-derivative** of a (i.e. $A(t) = \displaystyle\int_{t_0}^{t} a(s)ds$)
B is the anti-derivative of be^{-A} that cancels itself in 0.
Therefore, one needs to find both of the anti-derivatives in order to find the closed form.

Corollary : $\begin{cases} y'(t) = my(t) \\ y(0) = y_0 \in \mathbb{R} \end{cases}$ has one solution on $\mathbb{R}$: $t \mapsto y_0 e^{mt}$

Notes :

1. $y_0 = 0 \Longrightarrow y = 0$
2. if $m < 0$, a solution coming from $y_0 = 10^{-4}$ is really close to the solution coming from $y_0 = 0$ for $t > 0$: we will later call this a-stability
3. if $m > 0$, a solution coming from $y_0 = 10^{-4}$ is not close to the solution coming from $y_0 = 0$ for $t > 0$: we will later call this instability

3.2 Higher order $(p > 1)$ and scalar $(d = 1)$

p^{th} order linear ODE : ODE of the form $\displaystyle\sum_{i=0}^{p} a_i y^{(i)} = 0$ with :

- the a_i being real numbers
- $a_p \neq 0$ (we can set $a_p = 1$ with no loss of generality)

The initial condition will be given specifying all of the $y^{(i)}(0)$.

Let us consider such an equation.
The ODE is equivalent to $U' = AU$, with :

$$U = \begin{bmatrix} y \\ y' \\ \vdots \\ y^{(p-1)} \end{bmatrix}, U' = \begin{bmatrix} y' \\ y'' \\ \vdots \\ y^{(p)} \end{bmatrix}, A = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & 0 \\ 0 & 0 & 0 & 0 & 1 \\ -a_0 & -a_1 & -a_2 & \dots & -a_{p-1} \end{bmatrix}.$$

A^T is called the **companion matrix** to the ODE.
Therefore, **a pth order linear scalar ODE can be replaced by a first-order linear vectorial ODE** where the vector contains p components.

3.3 First order $(p = 1)$ and vectorial $(d > 1)$

Exponential of a matrix : let $A \in \mathcal{M}_d(\mathbb{R})$, $\exp(A) = \displaystyle\sum_{k=0}^{+\infty} \frac{1}{k!} A^k$

Solving a first-order linear vectorial ODE :
Duhamel's theorem :
1. $\forall A \in \mathcal{M}_d(\mathbb{R})$, $\forall u_0 \in \mathbb{R}^d$, $\begin{cases} u'(t) = Au(t) \\ u(0) = u_0 \end{cases}$ has a unique solution on $\mathbb{R}$: $t \mapsto u(t) = \exp(tA)u_0$

2. $\forall A \in \mathcal{M}_d(\mathbb{R}), \forall u_0 \in \mathbb{R}^d, \begin{cases} u'(t) = Au(t) + b(t) \\ u(0) = u_0 \end{cases}$ has a unique solution on $\mathbb{R}$: $t \mapsto u(t) = \exp(tA)u_0 + \int_0^t \exp((t-s)A)b(s)ds$

Computing $\exp(A)$:

$$\forall A \in \mathcal{M}_d(\mathbb{R}), \exists P \in \mathcal{GL}_p(\mathbb{R}), \exists J = D + N \in \mathcal{M}_d(\mathbb{R}), \begin{cases} A = P^{-1}JP \\ D \text{ is diagonal} \\ N \text{ is nilpotent} \end{cases}$$

J is called a **Jordan matrix**, and $\exp(J)$ can be computed easily thanks to the particular form of J.
To compute $\exp(A)$, we then use the formula $\exp(tA) = P^{-1}\exp(tJ)P$.

Corollary about stability :
Consider $u' = Au$, and denote $\sigma(A)$ the **spectrum** of A (i.e. the set of its eigenvalues).

1. $u(0) = 0 \Longrightarrow u(t) = 0$ is the **stationnary** solution
2. $\sigma(A) \subseteq \{z \in \mathbb{C}, \Re(z) < 0\} \Longrightarrow \begin{cases} u(t) \xrightarrow[t \to +\infty]{} 0 \\ ||u(t)|| \leq u(0)e^{-\lambda t}, \quad \lambda = |\max \sigma(A)| \end{cases}$
3. $\sigma(A) \cap \{z \in \mathbb{C}, \Re(z) > 0\} \neq \emptyset \Longrightarrow u(t)$ may diverge for some initial condition $u(0)$

4 Qualitative properties

4.1 Equilibrium of autonomous ODEs, Lyapunov Theorem, Linearization

For the following, we consider $t_0 \in \mathbb{R}$, $I \subseteq \mathbb{R}$ an open interval containing t_0, $\mathcal{U} \subseteq \mathbb{R}^d$ $(d \in \mathbb{N}^*)$ and $y_0 \in \mathcal{U}$.
Let $f : I \times \mathcal{U} \to \mathbb{R}^d$. We are interested in $\begin{cases} y'(t) = f(t, y(t)), \quad t \in I \\ y(t_0) = y_0 \end{cases}$

Autonomous ODE : an ODE is said to be autonomous if f is constant with respect to its first variable.
Example : $y'(t) = 3y(t)$ is autonomous $(f(t,x) = 3x)$.

We now consider an autonomous ODE using the same notations.
Equilibrium of an ODE : the ODE has an equilibrium at x^* if $f(x^*) = 0$.
When $y_0 = x^*$, the solution $y(t)$ is stationary.

Lyapunov stability : the equilibrium x^* is said to be Lyapunov stable if $\forall W \in \mathcal{V}(x^*), \exists V \in \mathcal{V}(x^*), y_0 \in V \Longrightarrow \forall t > 0, y(t) \in W$
In other words, Lyapunov stability of x^* means that the solutions starting "close enough" to x^* will remain close to x^*.
If an equilibrium x^* is not Lyapunov stable, it is called **unstable**.

Asymptotic stability : the equilibrium x^* is said to be asymptotically stable if $\exists V \in \mathcal{V}(x^*), y_0 \in V \Longrightarrow \lim_{t \to +\infty} y(t) = x^*$.
In other words, the asymptotic stability of an equilibrium means that the solution starting "close enough" to x^* will converge to x^*.

Exponential stability : the equilibrium x^* is said to be exponentially stable if it is asymptotically stable and $\exists \alpha, \beta > 0$, $V \in \mathcal{V}(x^*)$, $y_0 \in V \Longrightarrow \forall t > 0$, $||y(t) - x^*|| \leq \alpha ||y(0) - x^*|| e^{-\beta t}$.
In other words, the exponential stability of an equilibrium means that the solution starting "close enough" to x^* will converge *rapidly* to x^*.

Note : Exponential stability $\Longrightarrow$ Asymptotic stability $\Longrightarrow$ Lyapunov stability

Rewriting the corollary about stability :
Consider $u' = Au$, and denote $\sigma(A)$ the spectrum of A.
$x^* = 0$ is the only equilibrium.

1. $\sigma(A) \subseteq \{z \in \mathbb{C}, \Re(z) < 0\} \Longrightarrow 0$ is exponentially stable

2. $\sigma(A) \cap \{z \in \mathbb{C},\ \Re(z) > 0\} \neq \emptyset \Longrightarrow 0$ is unstable

Lyapunov theorem :
Assume that $f \in \mathcal{C}^2$, and consider x^* an equilibrium.
The **Jacobian matrix** of f is $Df(x) = (\frac{\partial f_i}{\partial x_j})_{ij}$.
1. The eigenvalues of the Jacobian matrix $Df(x^*)$ have negative real parts $\Longrightarrow x^*$ is exponentially stable.
2. At least one eigenvalue has a positive real part $\Longrightarrow x^*$ is unstable.

Linearization : Consider $y'(t) = f(y(t))$ with an equilibrium x^*, and $\xi(t) = y(t) - x^*$.
Then $\xi'(t) = Df(x^*)\xi(t)$ is the linearization of the ODE around x^*.
Intuition : Provided that $f \in \mathcal{C}^1$, the behavior of the ODE around x^* is "close" to the behavior of its linear counterpart.

Hyperbolic equilibrium : x^* is an hyperbolic equilibrium if all the eigenvalues of $Df(x^*)$ have non-zero real parts.
Otherwise, it is called **degenerate**.

4.2 Chaos

Chaotic behavior : the behavior of a system is chaotic when small differences in the initial conditions produce very different outcomes.
Consequently, long-term prediction of these systems is impossible in general, even though they are deterministic.

5 Numerical solutions

5.1 Introduction, the Euler Method

Euler (Forward) Method : the Euler forward method is given by :
$$\begin{cases} z_0 = y_0 \\ \forall n \in [\![0, N]\!],\ z_{n+1} = z_n + hf(t_n, z_n) \\ \forall n \in [\![0, N]\!],\ t_{n+1} = t_n + h \end{cases}$$
Note : it means that we substitute $y'(t_n)$ by $\dfrac{z_{n+1} - z_n}{h}$
Graphically, the Euler method means that we use the slope field to approximate the solution.

We will now try to estimate the error of the approximate solution.
To do so, let's introduce a few notations :

- $\forall n \in [\![0, N]\!],\ e_n = y(t_n) - z_n$
- $E_N = \max\limits_{0 \leq n \leq N} ||e_n||$

Question : do we have $E_N \xrightarrow[h_t \to 0]{} 0$?
If the answer is yes, the numerical method will be said **convergent**.

5.2 Discretization, k-steps Method

Mesh : let $N \in \mathbb{N}$.
The $N+1$-tuple $\mathcal{T} = (t_0, \ldots, t_N)$ where $0 = t_0 < t_1 < \cdots < t_N = T$ is called a mesh of $[0, T]$.
The positive number $h_n = t_{n+1} - t_n$ is called a **step**.
If h_n is constant, the mesh is called **regular**. The step is then often denoted h.

k-step method :
Let $d \in \mathbb{N}$, denote $||\cdot||$ the norm on $\mathbb{R}^d$.
Let $t_0 = 0$, $T > 0$.
Let $\mathcal{T} = (t_0, \ldots, t_N)$ be a mesh on $[0, T]$, and $f \in \mathcal{C}^0([0, T] \times \mathbb{R}^d)$, Lipschitz continuous with respect to its second variable.
We consider $\begin{cases} y'(t) = f(t, y(t)), \quad t \in I \\ y(t_0) = y_0 \end{cases}$

A sequence $(z_n)_{n\in[\![0,N]\!]}$, defined using $z_{n+1} = F_{\mathcal{T}}(t_n, z_{n+1}, z_n \dots, z_{n-k+1})$ such that z_n should approximate $y(t_n)$ is called a k-step method.

Explicit method : when the relation $z_{n+1} = F_{\mathcal{T}}(t_n, z_{n+1}, z_n \dots, z_{n-k+1})$ can be expressed $z_{n+1} = F_{\mathcal{T}}(t_n, z_n \dots, z_{n-k+1})$, the method will be called explicit : the value of z_{n+1} can be computed on the previous values of the sequence. An non-explicit method (i.e. a method that requires solving for z_{n+1}) is called an **implicit method**.

Note : while implicit method are usually more tricky to implement, they tend to have an advantage over explicit methods in terms of "stability" (defined later).

Increment function : when k = 1 and the mesh is regular with a step h, we often write $F_{\mathcal{T}}(t, x) = x + h\phi(t, h, x)$. The function ϕ defined on $[0, T] \times [0, h_t] \times \mathcal{U}$ is called the increment function.

5.3 Convergence

From now on, we will take $k = 1$, knowing that everything we do can easily be adapted to larger values of k.

Local Truncation Error :
Consider the IVP : $\begin{cases} y'(t) = f(t, y(t)) \\ y(t_0) = y_0 \end{cases}$, and a one-step method given by $\begin{cases} z_0 = y_0 \\ z_{n+1} = F_{\mathcal{T}}(t_n, z_{n+1}, z_n) \end{cases}$
The local truncation error is defined by $\varepsilon_n = y(t_{n+1}) - F_{\mathcal{T}}(t_n, y(t_{n+1}), y(t_n))$. It is a difference between the true solution and what would happen to it when we plugged it at t_n into the method.

Consistency : a numerical method is consistent if $\lim\limits_{h_t \to 0} \sum\limits_{n=0}^{N-1} ||\varepsilon_n|| = 0$.
Furthermore, the **consistency is of order p** if $\varepsilon_n = O((h_t)^{p+1})$.
Consistency means we can render the sum of the errors as small as we want by choosing a small enough step.

Necessary and sufficient condition for consistency : A one step explicit method given by Φ is consistent $\iff \forall t \in [0, T], \forall x \in \mathcal{U}, \Phi(t, 0, x) = f(t, x)$

Stability : let $(\eta_n)_{n\in\mathbb{N}}$ be a perturbing sequence.
For $z_0, w_0 \in \mathcal{U}$, define $\forall n \in [\![0, N-1]\!]$: $\begin{cases} z_{n+1} = F_{\mathcal{T}}(t_n, z_{n+1}, z_n) \\ w_{n+1} = F_{\mathcal{T}}(t_n, w_{n+1}, w_n) + \eta_n \end{cases}$
The numerical method is stable if $\exists K \in \mathbb{R}_+, \forall z_0, w_0, (\eta_n)_{n\in[\![0,N]\!]}, \quad \forall n \in [\![0, N]\!], ||z_n - w_n|| \leq K(||z_0 - w_0|| + \sum\limits_{n=0}^{N-1} ||\eta_n||)$
Stability means that the method does not amplify perturbations.

Useful lemma : $(\alpha_n)_{n\in\mathbb{N}}, (\beta_n)_{n\in\mathbb{N}}$ two non-negative sequences.
Let $q \in \mathbb{R}_+$, and assume $\forall n \in \mathbb{N}^*, \alpha_{n+1} \leq q\alpha_n + \beta_n$.
Then $\alpha_N \leq q_N \alpha_0 + \sum\limits_{n=0}^{N-1} q^{N-1-n} \beta_n$.

Stability of the one-step method : the one-step method is stable provided that ϕ is Lipschitz continuous in its second variable.
Note : it is the case under the hypothesis we chose for the Euler numerical method.

Convergence of a numerical method : the numerical method to approximate $\begin{cases} y'(t) = f(t, y(t)) \\ y(0) = y_0 \end{cases}$ converges if $\forall y_0 \in \mathcal{U}, \lim\limits_{h_t \to 0} \max\limits_{0 \leq n \leq N} ||y(t_n) - z_n|| = \lim\limits_{h_t \to 0} \max\limits_{0 \leq n \leq N} ||e_n|| = \lim\limits_{h_t \to 0} E_N = 0$
Furthermore, if $E_N = O((h_t)^p)$, the **method is of order p**.
Convergence means that the numerical solution approaches the exact solution of the ODE and converges to it as the step tends to zero.

Lax Equivalence : a consistent numerical scheme for which the initial-value problem is well-posed is convergent if and only if it is stable.

Note : as a corollary, the Euler numerical method is convergent.

A-stability : a method is called A-stable if $\lim(z_n) = 0$ when the method is applied with fixed positive h to any Dahlquist's test equation $\begin{cases} y'(t) = \lambda y(t), & \Re(\lambda) < 0 \\ y(t_0) = 1 \end{cases}$
Example : the Euler forward method is not A-stable.

5.4 A few good methods

Properties of the Euler forward method :

1. one-step explicit method
2. convergent of order one
3. not A-stable

RK2 (Runge-Kutta 2) :
Consider a one-step explicit method with increment function $\phi(t, h, x) = f(t + \frac{h}{2}, x + \frac{h}{2} f(x, t))$.
Starting with $\begin{cases} z_0 = y_0 \\ z_{n+1} = z_n + h_t f(t_n + \frac{h_t}{2}, z_n + \frac{h_t}{2} f(t_n, z_n)) \end{cases}$ is a way to compute the integro-differential form (see 2.2) with the mid-point rectangle method, called the Runge-Kutta 2 method.

Euler backward method : the Euler backward method is an implicit method which approximates y_{n+1} by solving numerically $y_{n+1} = y_n + hf(t_{n+1}, y_{n+1})$.
The Euler backward method is A-stable.

Crank-Nicolson Method : the Crank-Nicholson Method is given by
$\begin{cases} z_0 = y_0 \\ \forall n \in [\![0, N]\!],\ z_{n+1} = z_n + \frac{h_t}{2} f(t_n, z_n) + \frac{h_t}{2} f(t_{n+1}, z_{n+1}) \end{cases}$.
The Crank-Nicolson Method is a second order method. It corresponds to a trapezoidal method.

The Runge-Kutta Family of Methods : the family of explicit Runge-Kutta methods is given by $z_{n+1} = z_n + h \sum_{i=1}^{s} b_i k_i(t_n, z_n)$ where $\forall i \in [\![0, s]\!],\ k_i(t, x) = f(t + c_i h, x + \sum_{j=1}^{i-1} a_{ij} k_j(t, x))$
Parameters :

- Number of stages : s
- Runge-Kutta matrix : $\forall i \in [\![1, s]\!], \forall j \in [\![1, i-1]\!], a_{ij}$
- Runge-Kutta weights : $\forall i \in [\![1, s]\!], b_i$
- Runge-Kutta nodes : $\forall i \in [\![1, s]\!], c_i$

The RK4 method is a fourth-order method.

PDE - Modeling with PDEs and Classification

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Introduction

Partial differential equation : equation that relates a function with its derivatives (with respect to d variables). The unknown is a function.
Example : $\dfrac{\partial u}{\partial t} - \Delta u = f$

1 Differential calculus

1.1 The outward normal vector field

Let $d \in \mathbb{R}^d$ and $\Omega \subseteq \mathbb{R}^d$.
Boundary : the boundary of Ω is $\partial\Omega = \overline{\Omega} \setminus \mathring{\Omega}$.

Regular open set of class $\mathcal{C}^1$: Ω is a regular open set of class $\mathcal{C}^1$ if $\forall x_0 \in \partial\Omega$, $\exists r > 0, \exists \gamma \in \mathcal{C}^1(\mathbb{R}^{d-1}, \mathbb{R}^d)$, $\Omega \cap B(x_0, r) = \{x \in B(x_0, r),\ x_d > \gamma(x_1, \dots, x_{d-1}\}$ (after reorienting the coordinates axes if necessary).
In other words, the boundary of Ω is locally the graph of a $\mathcal{C}^1$ function.

Outward unit normal vector field : let Ω be a regular open set of class $\mathcal{C}^1$.
The outward unit normal vector field to Ω is defined for each point $x \in \partial\Omega$ as the unit vector normal to the tangent plane to Ω in x pointing towards the exterior.
It is often denoted by $n(x) = \begin{bmatrix} n_1(x) \\ \vdots \\ n_d(x) \end{bmatrix}$.
Note : this is the vector according to which we orient surfaces in physics !

1.2 Partial derivatives

Let $d, N \in \mathbb{N}^*$, and $U \subseteq \mathbb{R}^d$ be a non-empty *open* set.

Differentiability : the function $f : U \to \mathbb{R}^N$ is differentiable if $\exists L \in \mathcal{L}(\mathbb{R}^d, \mathbb{R}^{\mathbb{N}})$, $\exists V \in \mathcal{V}(x_0)$, $\forall x \in V$, $f(x) - f(x_0) = L(x - x_0) + o(||x - x_0||)$.

Partial derivative : let $B = (e_1, \dots, e_n)$ be a basis of $\mathbb{R}^d$.
$\forall i \in [\![1, d]\!]$, $\dfrac{\partial f}{\partial x_i}(x_0) = \lim\limits_{h \in \mathbb{R}, h \to 0} \dfrac{f(x_0 + he_i) - f(x_0)}{h}$ is the partial derivative of f with respect to x_i in x_0.
It is also denoted as $\partial_{x_i} f(x_0)$.

Differential : the differential of $f = \begin{bmatrix} f_1 \\ \vdots \\ f_N \end{bmatrix}$ in x_0 is $Df(x_0) = \begin{bmatrix} \partial_{x_1} f_1(x_0) & \dots & \partial_{x_d} f_1(x_0) \\ \vdots & \ddots & \vdots \\ \partial_{x_1} f_N(x_0) & \dots & \partial_{x_d} f_N(x_0) \end{bmatrix}$

Gradient : ***in the case*** $N = 1$, f is a scalar valued function and L is a linear form.
The gradient of f at x_0, denoted by $\nabla f(x_0)$ or $\mathrm{grad} f(x_0)$, is the vector such that $\forall h \in \mathbb{R}^d, L(h) = \langle \nabla f(x_0), h \rangle_{\mathbb{R}^d}$.
We can also write the expression of the gradient : $\nabla f(x_0) = \begin{bmatrix} \partial_{x_1} f(x_0) \\ \vdots \\ \partial_{x_d} f(x_0) \end{bmatrix}_B$.

Divergence : ***in the case*** $N = d$, L is an endomorphism of $\mathbb{R}^d$.
The divergence of f at x_0, denoted by $\nabla \cdot f(x_0)$ or $\mathrm{div} f(x_0)$, is the trace of $Df(x_0)$: $\nabla \cdot f(x_0) = \sum_{1 \leq i \leq d} \partial_{x_i} f_i(x_0)$ in B.
Interpretation of the divergence : the divergence of a vector field measures its tendency to close in toward a point or to repel from it.

Laplace operator/Laplacian : differential operator defined by the divergence of the gradient : $\Delta f(x_0) = \nabla \cdot (\nabla f)(x_0) = \sum_{i=1}^{d} \frac{\partial^2 f}{\partial x_i^2}(x_0)$.
It is denoted by Δ or ∇^2.

Multi-index notation : a d-dimensional multi-index is an element of $\mathbb{N}^d$.
The sum of the components of a multi-index $\alpha = (\alpha_1, \dots, \alpha_d)$ is usually denoted $|\alpha|$.

Higher-order partial derivative : let $\alpha = (\alpha_1, \dots, \alpha_d)$ be a d-dimensional multi-index.
$D^\alpha = \frac{\partial^{|\alpha|}}{\partial x_1^{\alpha_1} \dots \partial x_d^{\alpha_d}}$ is the derivative with respect the to multi-index α.
Linear differential operator : a linear differential operator P of order k is a linear combination of derivatives $P = \sum_{|\alpha| \leq k} a_\alpha D^\alpha$ where each α is a multi-index and each a_α is a function, and at least one a_α is non zero for $|\alpha| = k$.
The polynomial $\sum_{|\alpha| \leq k} a_\alpha(x) X^\alpha$ is called the **symbol of the operator**, where $X^\alpha = X_1^{\alpha_1} \dots X_d^{\alpha_d}$.

1.3 Integration by parts

Support of a function : let f be a function defined from an open interval $I \subseteq \mathbb{R}$ to $\mathbb{R}$.
The support of f is the closure of the set of points where f is non zero : $\mathrm{supp}(f) = \overline{\{x \in I, f(x) \neq 0\}}$.

Green's Formula : let $u \in \mathcal{C}^1$ with compact support $\overline{\Omega}$
$\forall i \in [\![1, d]\!]$, $\int_\Omega \frac{\partial u}{\partial x_i}(x) dx = \int_{\partial\Omega} u(s) n_i(s) ds$

Note : if $d = 1$, $\Omega =]a, b[$ with $a < b$, this property can be written as : $\int_\Omega \frac{du}{dx}(x) dx = u(b) - u(a)$.

Integration by parts (1) : let $u, v \in \mathcal{C}^1(\overline{\Omega})$.
$\forall i \in [\![1, d]\!]$, $\int_\Omega \frac{\partial u}{\partial x_i}(x) v(x) d\lambda^{(d)} = -\int_\Omega u(x) \frac{\partial v}{\partial x_i}(x) d\lambda^{(d)} + \int_{\partial\Omega} u v n_i d\sigma$, where :

- $\lambda^{(d)}$ is the Lebesgue measure on $\mathbb{R}^d$
- $\int_{\partial\Omega} u v n_i d\sigma$ is the integral on the boundary $\partial\Omega$ of $u \times v \times n_i$

Integration by parts (2) : let $u \in \mathcal{C}^2(\overline{\Omega})$, $v \in \mathcal{C}^1(\overline{\Omega})$ with compact support in $\overline{\Omega}$. Then :
$\int_\Omega (\Delta u) v = -\int_\Omega \nabla u \cdot \nabla v + \int_{\partial\Omega} v \nabla u \cdot n$
$\frac{\partial u}{\partial n} = \nabla u \cdot n$ is called the **normal derivative**.

2 Definitions

2.1 Types of PDEs

Order of a PDE : the order of a PDE is determined by the term with the highest derivatives.

Linearity : a PDE is linear if it is linear in the unknown function and its derivatives.

Classification of PDEs : second order linear partial differential equations (PDEs) can be classified into three categories :

1. Elliptic
2. Parabolic
3. Hyperbolic

When $d = 2$, the general form of such a PDE is : $A\dfrac{\partial^2 u}{\partial x_1^2} + B\dfrac{\partial^2 u}{\partial x_1 x_2} + C\dfrac{\partial^2 u}{\partial x_2^2} + D\dfrac{\partial u}{\partial x_1} + E\dfrac{\partial u}{\partial x_2} + Fu + G = 0.$
The PDE is said to be :

- **elliptic** if $B^2 - 4AC < 0$
- **parabolic** if $B^2 - 4AC = 0$ and $(A, B, C) \neq (0, 0, 0)$
- **hyperbolic** if $B^2 - 4AC > 0$ or $(A, B, C) = (0, 0, 0)$

Note : $AX^2 + BXY + CY^2 + DX + EY + F = 0$ is the equation of a conic :

- an **ellipse** if $B^2 - 4AC < 0$
- a **parabola** if $B^2 - 4AC = 0$ and $(A, B, C) \neq (0, 0, 0)$
- an **hyperbola** if f $B^2 - 4AC > 0$

We call a PDE an **evolution equation** when it models the development in time of a system.
On the contrary, we talk of **steady state** when the behavior of the system does not change over time.

2.2 Wellposedness

Boundary conditions (B.C.) : let $\Omega \subseteq \mathbb{R}^d$ be a regular open set of class $\mathcal{C}^1$.
Consider a PDE $\mathcal{A}u = f$ on Ω, where $\mathcal{A}$ is the differential operator, u is the unknown and f is the data.
The boundary condition is an equation that u must satisfy on $\partial\Omega$ (or a part of it).

Dirichlet boundary condition : $u = g$ on $\partial\Omega$ for a given g.
The Dirichlet boundary condition is said to be **homogeneous** if $g = 0$, i.e. $u = 0$ on $\partial\Omega$.

Neumann boundary condition : $\dfrac{\partial u}{\partial n} = g$ on $\partial\Omega$ for a given g.
The Neumann boundary condition is said to be **homogeneous** if $g = 0$, i.e. $\dfrac{\partial u}{\partial n} = 0$ on $\partial\Omega$.

Wellposedness in the sense of Hadamard : let E, F be two spaces.
Let $f \in F$ be a data and $\mathcal{A} : E \to F$.
We are looking for solutions $u \in E$ to $\mathcal{A}(u) = f$.
The PDE is well-posed in the sense of Hadamard if :

1. a solution u exists
2. the solution u is unique
3. u changes "continuously" to respect to f

A PDE that is not well-posed is said to be **ill-posed**.

PDE - Distributions

Tom Jeandé (from John Cagnol's videos)

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Introduction

Origin of distributions : distributions are, in a way, the equivalent for functions to what complex numbers are to real numbers.
Complex numbers were invented while looking for a solution to $x^2 = -1$; similarly, distributions were invented while looking for the derivative of a non-derivable function.

Note : the following results will be established in $\mathbb{R}$, but can be extended to $\mathbb{R}^d$

1 Definition of a distribution

1.1 The set of distributions $\mathcal{D}'$

Denote $C_0^0(\mathcal{I})$ or $C_c^0(\mathcal{I})$ the set of $\mathbb{R}$-valued continuous functions on $\mathcal{I}$ with compact support.
Note : f has compact support on $(a, b) \Longrightarrow f(x) \xrightarrow[x \to a \text{ or } b]{} 0$

Denote $L^1_{\text{loc}}(\mathcal{I})$ the set of locally integrable functions on $\mathcal{I}$, i.e. functions f such that $\forall K \subseteq \mathcal{I}$ compact, $\displaystyle\int_K |f(x)|dx < +\infty$.

Bump functions : the set of bump functions $\mathcal{D}(\mathcal{I})$ is given by $\mathcal{D}(\mathcal{I}) = C_0^0(\mathcal{I}) \cap C^\infty(\mathcal{I})$.
In other words, bump functions are functions whose support is bounded which can be differentiated for all degrees of differentiation.
$\mathcal{D}(\mathcal{I})$ is also denoted by $C_c^\infty(\mathcal{I})$ or $C_0^\infty(\mathcal{I})$.
$\mathcal{D}(\mathcal{I})$ is called the **test function space**. It is a **real vector space**.

Convergence in the test function space : let $(\phi_n)_{n\in\mathbb{N}} \in \mathcal{D}(\mathcal{I})^{\mathbb{N}}$ and $\phi \in \mathcal{D}(\mathcal{I})$.
ϕ_n converges to ϕ if :

1. $\exists K \subseteq \mathcal{I}$ compact, $\forall n \in \mathbb{N}$, $\text{supp}(\phi_n) \subseteq K$
2. $\forall m \in \mathbb{N}, \phi_n^{(m)} \to \phi^{(m)}$ uniformly in $\mathcal{I}$
   (i.e. $\forall m \in \mathbb{N}$, $\forall \varepsilon > 0$, $\exists N \in \mathbb{N}$, $\forall n \geq N$, $||\phi_n^{(m)} - \phi^{(m)}||_\infty \leq \varepsilon$)

We will note $\phi_n \xrightarrow{\mathcal{D}(\mathcal{I})} \phi$ or $\phi = \lim_{n\to\infty}^{\mathcal{D}(\mathcal{I})} \phi_n$.

Distribution space : the space of distribution is the topological dual of $\mathcal{D}(\mathcal{I})$.
It is denoted by $\mathcal{D}'(\mathcal{I})$, and its elements are called **distributions/generalized functions**.
In other words, a distribution is a **continuous linear** function $T : \mathcal{D}(\mathcal{I}) \to \mathbb{R}$, which means that :

1. $T(\varphi + \lambda\psi) = T(\varphi) + \lambda T(\psi)$ (linearity)
2. $\phi_n \xrightarrow{\mathcal{D}(\mathcal{I})} \phi \Longrightarrow \langle T, \phi_n\rangle \xrightarrow{\mathbb{R}} \langle T, \phi\rangle$ (continuity, see notation below)

Notation : $\langle T, \phi\rangle = T(\phi)$

Dirac distribution in $a \in \mathcal{I}$: δ_a such that $\forall \phi \in \mathcal{D}(\mathcal{I})$, $\langle \delta_a, \phi\rangle = \phi(a)$

1.2 Regular distributions

Regular distribution : the regular distribution associated to $f \in L^1_{\text{loc}}(\mathcal{I})$ is defined by $\mathcal{T}_f : \phi \mapsto \displaystyle\int_I f(x)\phi(x)dx$
Note : not all distributions are regular distribution (for example the Dirac distribution)

Injectivity of $f \mapsto \mathcal{T}_f$: $\mathcal{T} : f \mapsto \mathcal{T}_f$ is injective from $L^1_{\text{loc}}(\mathcal{I})$ to $\mathcal{D}(\mathcal{I})$, meaning that all functions of L^1_{loc} can be represented by their regular distribution $\mathcal{T}_f$.
Consequence : we can identify $L^1_{\text{loc}}(\mathcal{I})$ to a subset of $\mathcal{D}'(\mathcal{I})$ (the set of regular distributions), and write : $L^1_{\text{loc}}(\mathcal{I}) \subseteq \mathcal{D}'(\mathcal{I})$.
Notation : $\langle f, \varphi\rangle = \langle \mathcal{T}_f, \varphi\rangle$

2 Operations in $\mathcal{D}'$

2.1 Sum and product

Sum of two distributions : let T_1, T_2 be two distributions.
$T_1 + T_2$, defined by $\forall \varphi \in \mathcal{D}(\mathcal{I})$, $\langle T_1 + T_2, \varphi\rangle = \langle T_1, \varphi\rangle + \langle T_2, \varphi\rangle$, **is a distribution**.
This property extends to the sum of $L^1_{\text{loc}}(\mathcal{I})$ since $\mathcal{T}_{f_1+f_2} = \mathcal{T}_{f_1} + \mathcal{T}_{f_2}$.

Note : we cannot define the product of two distributions this way, as we cannot prove linearity.

Product of a distribution and a smooth function : let $T \in \mathcal{D}'(\mathcal{I})$ and $h \in C^\infty(\mathcal{I})$.
hT, defined by $\forall \varphi \in \mathcal{D}(\mathcal{I})$, $\langle hT, \varphi\rangle = \langle T, h\varphi\rangle$ **is a distribution**.

2.2 Derivative

Derivative of a distribution : let $T \in \mathcal{D}'(\mathcal{I})$.
T', defined by $\forall \varphi \in \mathcal{D}(\mathcal{I})$, $\langle T', \varphi\rangle = -\langle T, \varphi'\rangle$, **is a distribution**, known as the derivative of T.
This is an extension of the differentiation of differentiable functions of $L^1_{\text{loc}}(\mathcal{I})$ (proof uses an integration by parts + compact support of φ).

Useful example : derivative of the distribution associated to the Heaviside distribution $H_0 = \mathbb{1}_{\mathbb{R}_+} \in L^1_{\text{loc}}(\mathbb{R})$
$\forall \varphi \in \mathcal{D}(\mathbb{R})$, $\langle T'_{H_0}, \varphi\rangle = -\displaystyle\int_0^{+\infty} \varphi'(x)dx = -\lim_{x\to+\infty} \varphi(x) + \varphi(0) = \varphi(0)$. Thus, $T'_{H_0} = \delta_0$.

Note : (a, b) is the English notation for $]a, b[$ in French.
Extension to $L^1_{\text{loc}}(\mathcal{I})$: let $a, b \in \overline{\mathbb{R}}$ such that $a < b$.
If :

1. $f \in L^1_{\text{loc}}(a, b)$ is differentiable on (a, b)
2. $f' \in L^1_{\text{loc}}(a, b)$

Then $\mathcal{T}_{f'} = \mathcal{T}'_f$.

Jumps formula : let $\mathcal{I} = (a_0, a_{k+1})$, and $f \in C^1_{\text{piecewise}}(\mathcal{I})$.
Let $a_1, \ldots, a_k$ be the points where f is not continuous, and denote f' the derivative of the restriction of f to each sub-interval (a_i, a_{i+1}), for $i \in [\![0, k]\!]$. Then :

$\mathcal{T}_f' = \mathcal{T}_{f'} + \sum_{i=1}^{k} (f(a_i^+) - f(a_i^-))\delta_{a_i}$.

Note : with distributions, we lose the ability to compute a "function" at a given point.

3 Sobolev spaces

3.1 Definition of H^k and regularity

Reminder : $f \in L^2(\mathcal{I})$ (f is a function) $\iff \int_{\mathcal{I}} |f|^2 < +\infty$

Note : to say that ***a distribution*** T **is in** $L^2(\mathcal{I})$ means that $\exists f \in L^2(\mathcal{I})$, $\forall \varphi \in \mathcal{D}(\mathcal{I})$, $\langle T, \varphi \rangle = \int_{\mathcal{I}} f(x)\varphi(x)dx$

Sobolev space of order 1 : the Sobolev space of order 1 on $\mathcal{I}$, denoted $H^1(\mathcal{I})$ is defined by :
$H^1(\mathcal{I}) = \{v \in L^2(\mathcal{I}),\ \mathcal{T}_v' \in L^2(\mathcal{I})\} = \{v \in L^2(\mathcal{I}),\ v' \in L^2(\mathcal{I})\}$, where v' is the distributional derivative of v.
If $\mathcal{I} = (a, b)$ is a bounded interval, we note $H^1(\mathcal{I}) = H^1(a, b)$.

The $H^1(\mathcal{I})$ Hilbert space :
The space $H^1(\mathcal{I})$ endowed with the inner product $\langle \cdot\,, \cdot \rangle_{H^1} : (u, v) \mapsto \langle u, v \rangle_{L^2} + \langle u', v' \rangle_{L^2}$ is complete for the norm $||\cdot||_{H^1(\mathcal{I})} : v \mapsto \sqrt{||v||^2_{L^2(\mathcal{I})} + ||v'||^2_{L^2(\mathcal{I})}}$.
It is a Hilbert space.

Sobolev spaces of higher order : let $k \in \mathbb{N}$.
$H^k(\mathcal{I}) = \{u \in L^2(\mathcal{I}),\ \forall m \in [\![0, k]\!],\ u^{(m)} \in L^2(\mathcal{I})\}$ is the Sobolev space of order k.
$H^k(\mathcal{I})$ endowed with the inner product $(u, v) \mapsto \sum_{0 \le m \le k} \langle u^{(m)}, v^{(m)} \rangle_{L^2(\mathcal{I})}$ is a Hilbert space.

We note $H^0(\mathcal{I}) = L^2(\mathcal{I})$.

Basis function : $f : x \mapsto \begin{cases} x \text{ if } x \in (0, 1) \\ 2 - x \text{ if } x \in [1, 2) \end{cases}$ defined on $\mathcal{I} = (0, 2)$.
$f \in H^1(0, 2)$ but $f \notin H^2(0, 2)$.

3.2 Trace theorem and integration by parts

Rellich theorem : let $\mathcal{I} = (a, b)$ be a bounded open interval of $\mathbb{R}$.
$\forall u \in H^1(a, b)$, $\exists \tilde{u}$ in the class of u such that :

1. $\tilde{u}$ is continuous on (a, b)
2. $\tilde{u}$ is an anti-derivative of u' (i.e. $\forall x, y \in [a, b]$, $\tilde{u}(x) - \tilde{u}(y) = \int_{[y,x]} u'(t)dt$)

Furthermore :

1. $\exists C > 0$ depending only on $b - a$ such that $\forall u \in H^1(a, b)$, $||\tilde{u}||_\infty \le C||u||_{H^1}$
2. For any bounded sequence of elements of $H^1(a, b)$, one can extract a subsequence that converges in $C^0([a, b])$

Since $H^1(a, b) \subseteq C^0([a, b])$, one can evaluate u at a and b.
Trace of $u \in H^1(a, b)$: $\forall u \in H^1(a, b)$, the trace of u is the couple $(u(a), u(b))$.
The linear mapping $\gamma_0 : u \mapsto (u(a), u(b))$ is called the **trace operator**.

Continuity of the trace operator :
$\exists C > 0$ depending only on $b - a$ such that $\forall x \in [a, b]$, $\forall u \in H^1(a, b)$, $|u(x)| \le C||u||_{H^1}$.
Subsequently, γ_0 is continuous from $H^1(a, b)$ to $\mathbb{R}^2$.

Integration by parts : let $u, v \in H^1(a, b)$. Then :
$\int_{(a,b)} uv' = -\int_{(a,b)} vu' + u(b)v(b) - u(a)v(a)$

Integration by parts 2 : let $u \in H^2(a,b)$ and $v \in H^1(a,b)$. Then :
$\int_{(a,b)} u'v' + \int_{(a,b)} u''v = u'(b)v(b) - u'(a)v(a)$

3.3 The space H_0^1

Note : let $\mathcal{I}$ be a bounded set of $\mathbb{R}$.
$\mathcal{D}(\mathcal{I})$ is dense in $L^2(\mathcal{I})$, but not in $H^1(\mathcal{I})$! (even though $H^1(\mathcal{I}) \subseteq L^2(\mathcal{I})$)
This is possible because we consider different topologies in both cases.

$H_0^1(\mathcal{I}) = \gamma_0^{-1}(\{(0,0)\})$

Properties of $H_0^1(\mathcal{I})$:

1. $H_0^1(\mathcal{I}) = \overline{\mathcal{D}(\mathcal{I})}$ for the norm of $H^1(\mathcal{I})$
2. $H_0^1(\mathcal{I}) \subsetneq H^1(\mathcal{I})$ (strict inclusion)
3. $H_0^1(\mathcal{I})$ endowed with the norm of $H^1(\mathcal{I})$ is a Hilbert space

Poincaré/Friedrichs theorem : let $\mathcal{I}$ be a bounded interval of $\mathbb{R}$.
Then $\exists C_{\mathcal{I}}$ depending only on $\mathcal{I}$ such that $\forall v \in H_0^1(\mathcal{I})$, $||v||_{L^2(\mathcal{I})} \leq C_{\mathcal{I}}||v'||_{L^2(\mathcal{I})}$

Consequence : the semi-norm $v \mapsto ||v'||_{L^2(\mathcal{I})}$ defined on $H^1(\mathcal{I})$ by $v \mapsto |v|_{H^1(\mathcal{I})} = ||v'||_{L^2(\mathcal{I})}$ satisfies in $H_0^1(\mathcal{I})$
$\forall v \in H_0^1(\mathcal{I}), |v|_{H^1(\mathcal{I})} \leq ||v||_{H^1(\mathcal{I})} \leq \sqrt{1 + C_{\mathcal{I}}^2}\, |v|_{H^1(\mathcal{I})}$

Norm on $H_0^1(\mathcal{I})$: $||\cdot||_{H_0^1(\mathcal{I})} : v \mapsto |v|_{H^1(\mathcal{I})}$ is a norm on $H_0^1(\mathcal{I})$.
Inner product in $H_0^1(\mathcal{I})$: $\langle \cdot\,, \cdot \rangle_{H_0^1(\mathcal{I})} : (u,v) \mapsto \int_{\mathcal{I}} u'v'$ is an inner product on $H_0^1(\mathcal{I})$.
The space $H_0^1(\mathcal{I})$ endowed with the inner product $\langle \cdot\,, \cdot \rangle_{H_0^1(\mathcal{I})}$ is a Hilbert space.

PDE - Distributions

Tom Jeandé (from John Cagnol's videos)

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Introduction

Example : we consider the following problem :
Let Ω be a regular open set of class $\mathcal{C}^1$.

$$\mathcal{P} : \begin{cases} -\Delta \forall x \in \Omega,\ u(x) + c(x)u(x) = f(x) \\ u|_{\partial\Omega} = 0 \end{cases}$$

Dirichlet problem : the Dirichlet problem consists in finding a function u which is :

1. Solution to a PDE on the interior of the domain Ω
2. Taking prescribed values on the boundary of Ω

Classical solution : for the above equation, if $u \in \mathcal{C}^2(\Omega) \cap \mathcal{C}^0(\overline{\Omega})$, the solution is called classical. Otherwise, the solution is called **weak/variational**.

Goal : in this chapter, our goal is to established the well-posedness of this equation.

1 Duality in Hilbert Spaces

Let $\mathcal{I}$ be an open interval of $\mathbb{R}$: $(a,b), (-\infty, b)$ or $(a, +\infty)$.

Reminder from CIP : $L^2(E, \mathcal{E}, \mu)$ endowed with $\langle \cdot, \cdot \rangle : (f,g) \mapsto \int_E fg d\mu$ is a Hilbert space on the field $\mathbb{R}$.

Riesz representation theorem : let $\mathcal{H}$ be a Hilbert space, and $u \in \mathcal{H}'$.
Then $\exists! x_u \in \mathcal{H}$, $\forall x \in \mathcal{H}$, $u(x) = \langle x, x_u \rangle$.

Consequence in L^2 : $\forall u \in (L^2(\mathcal{I}))'$, $\exists! g_u \in L^2(\mathcal{I})$, $\forall f \in L^2(\mathcal{I})$, $u(f) = \langle f, g_u \rangle$.
We can thus *identify* $L^2(\mathcal{I})$ with its dual : $L^2(\mathcal{I}) = (L^2(\mathcal{I}))'$.

Pivot space : from the previous theorem, $V \subseteq L^2(\mathcal{I}) \Longrightarrow (L^2(\mathcal{I}))' \subseteq V'$.
Therefore we have $V \subseteq L^2(\mathcal{I}) \subseteq V'$, and $L^2(\mathcal{I})$ is called the pivot space.

2 The Lax-Milgram Theorem

Coercivity : let $\mathcal{H}$ be a Hilbert space.
Let $a : \mathcal{H} \times \mathcal{H} \to \mathbb{R}$ be a bilinear form.
a is coercive/$\mathcal{H}$-elliptic if $\exists \alpha > 0$, $\forall x \in \mathcal{H}$, $a(x,x) \geq \alpha ||x||^2$.
Notes :

1. Since the norm is dependent on the Hilbert space, coercivity depends on the Hilbert space
2. If a is coercive, $||x|| \to +\infty \Longrightarrow a(x,x) \to +\infty$

Continuity of a bilinear form : let $\mathcal{H}$ be a Hilbert space.
$a : \mathcal{H} \times \mathcal{H} \to \mathbb{R}$ is continuous if $\exists C > 0, \forall x, y \in \mathcal{H}$, $a(x,y) \leq \alpha ||x|| \, ||y||$
Note : linearity implies continuity *in finite-dimensional spaces*. In infinite-dimensional spaces, it has to be proven !

Lax-Milgram theorem : let $\mathcal{H}$ be a Hilbert space.
Consider a a continuous and coercive bilinear form, and $f \in \mathcal{H}'$.
Then, consider the equation $\forall u \in \mathcal{H}$, $a(x,u) = f(u)$:

1. It has one and only one solution x
2. The application $f \mapsto x$ is linear and continuous from $\mathcal{H}'$ to $\mathcal{H}$

3 Wellposedness of Elliptic Problems

3.1 $-u'' = f$

Consider the following problem :
$f \in L^2(0,1)$, $\begin{cases} -u'' = f \text{ in } (0,1) \\ u(0) = u(1) = 0 \end{cases}$
Our goal is to prove the wellposedness of this equation.

3.1.1 Find the weak formulation

Assume that $u \in \mathcal{C}^2([0,1])$, $u(0) = u(1) = 0$ is a solution.
Let $\phi \in \mathcal{C}_c^1(0,1)$.
Using $-u'' = f$, we find that $-\int_{(0,1)} u''\phi = \int_{(0,1)} f\phi$.
Integrating by parts : $\int_{(0,1)} u'\phi' = \int_{(0,1)} f\phi$ (since $\phi(0) = \phi(1) = 0$).

3.1.2 The variational formulation

Define :

1. A bilinear form $a : (u, v) \mapsto \int_{(0,1)} u'v'$
2. A linear form $\ell : v \mapsto \int_{(0,1)} fv$

Since u vanishes on the boundary, let's choose $H = H_0^1(0,1)$ endowed with $||\cdot||_{H_0^1}$.
a, ℓ are defined on $H \subseteq H^1(0,1)$.

Variational formulation of the problem : find $u \in H$, $\forall v \in H$, $a(u,v) = \ell(v)$.

3.1.3 Continuity of a and ℓ

Our goal is to apply the Lax-Milgram Theorem.

Let $(u,v) \in (H_0^1(0,1))^2$.
Using Cauchy-Schwarz, $|a(u,v)| = \left|\int_{(0,1)} u'v'\right| \leq ||u'||_{L^2}||v'||_{L^2} \leq ||u||_{H_0^1}||v||_{H_0^1}$.
Using Poincaré, $|\ell(v)| = \left|\int_{(0,1)} fv\right| \leq ||f||_{L^2}||v||_{L^2} \leq C_\Omega||f||_{L^2}||v||_{H_0^1}$.
Thus, a, ℓ are continuous on $H_0^1(0,1)$.

3.1.4 Coercivity of a

Let $u \in H_0^1(0,1)$.
Then $a(u,u) = \int_0^1 u'^2 = ||u||_{H_0^1}^2$.
Therefore, a is coercive.

Note : the tricky part is often to chose the right Hilbert space for a to be coercive.

3.1.5 Existence and uniqueness of the weak solution

Using the Lax-Milgram theorem on our problem, we find that there exists one and only one $u \in H$, $\forall v \in H$, $a(u,v) = \ell(v)$.
Furthermore, $a(u,u) = \ell(u) \Longrightarrow ||u||_{H_0^1} \leq C_\Omega||f||_{L^2}$.
The variational formulation is thus well-posed.

3.1.6 Solving the PDE in D'

Since $\mathcal{D}(0,1) \subseteq H_0^1(0,1)$, $\forall \phi \in \mathcal{D}(0,1)$, $\int_0^1 u'\phi' = \int_0^1 f\phi$, which is equivalent to $\forall \phi \in \mathcal{D}(0,1)$, $\langle u', \phi' \rangle = \langle f, \phi \rangle$.
This is, again, equivalent to $\forall \phi \in \mathcal{D}(0,1)$, $\langle -u'' - f, \phi \rangle = 0$.
Therefore, $-u'' = f$ in $\mathcal{D}'(0,1)$, and $u(0) = u(1) = 0$.

3.1.7 Regularity of the solution to the PDE

We have $u, u' \in L^2(0,1)$ and $-u'' = f \in L^2(0,1)$, so $u'' \in L^2(0,1)$.
Therefore, $u \in H^2(0,1)$.
Poincaré inequality : $||u||_{H^2} = \sqrt{||u||_{H^1}^2 + ||u''||_{L^2}^2} \leq \sqrt{(1 + C_\Omega^2)||u||_{H_0^1}^2 + ||f||_{L^2}^2} \leq \sqrt{2 + C_\Omega^2}C_\Omega||f||_{L^2}$.

Note : if $f \in C^0([0,1]) \cap L^2(0,1)$, then $u \in C^2([0,1])$ is a classical solution to the problem.

3.1.8 Conclusion

Let $f \in L^2(0,1)$.

The problem : find a solution $u \in H^2(0,1) \cap H_0^1(0,1)$ such that $\begin{cases} -u'' = f \text{ in } (0,1) \\ u(0) = u(1) = 0 \end{cases}$ is well-posed, since there exists a constant $C_\Omega > 0$ such that for a given $f \in L^2(\Omega)$, there exists a unique solution $u_f \in H^2(0,1) \cap H_0^1(0,1)$ continuously dependent on f : $||u_f||_{H^2} \leq C_\Omega ||f||_{L^2}$.

3.1.9 Recap of the seven steps

The seven steps are the following :

1. Find the weak formulation
2. Write the variational formulation
3. Prove the continuity of a and ℓ
4. Prove the coercivity of a
5. Apply the Lax-Milgram theorem
6. Get a solution in the sense of distributions
7. Get the regularity of the solution

Maximum principle

3.2 $-\Delta u = f$

The method used in section 3.1 can be adapted for higher dimensions.
It requires defining distributions $\mathcal{D}(\Omega)$ with $\Omega \subseteq \mathbb{R}^d$.
It also requires defining Sobolev spaces in higher dimensions.

Example : let $f \in L^2(\Omega)$, we are interested in solving $\begin{cases} -\Delta u = f \text{ in } \Omega \\ u|_{\partial\Omega} = 0 \end{cases}$

Note : it cannot be solved explicitly in general.

Existence and uniqueness of a solution : let $\Omega \subseteq \mathbb{R}^d$ be a open bounded set, and $f \in L^2(\Omega)$.

1. There exists a unique solution $u \in H_0^1(\Omega)$ to the variational formulation associated to the Dirichlet problem.
   Furhtermore, u satisfies $-\Delta u = f$ almost everywhere in Ω, and $u \in H_0^1(\Omega)$.
   Finally, there exists C_Ω independent of f such that $||u||_{H^1}(\Omega) \leq C_\Omega ||f||_{L^2(\Omega)}$.
2. If Ω is a regular bounded open set of class C^1, then u is solution to the Dirichlet problem $-\Delta u = f$ almost everywhere and $u = 0$ almost everywhere on $\partial\Omega$.

Let's prove that theorem using the seven steps :

3.2.1 Find the weak formulation

Assume $u \in C^2(\Omega)$, $u|_{\partial\Omega} = 0$ is a solution.
Let $\phi \in C_c^1(\Omega)$. Then $-\displaystyle\int_\Omega (\Delta u)\phi = \int_\Omega f\phi$.
Green's formula gives : $\displaystyle\int_\Omega \nabla u \cdot \nabla\phi - \int_\Omega \phi \nabla u \cdot n = \int_\Omega f\phi$.
Since $\phi|_{\partial\Omega} = 0$, we get : $\displaystyle\int_\Omega \nabla u \cdot \nabla\phi = \int_\Omega f\phi$.

Reminder : Green's formula :
Let $u \in \mathcal{C}^1$ with compact support $\overline{\Omega}$
$\forall i \in [\![1, d]\!]$, $\displaystyle\int_\Omega \frac{\partial u}{\partial x_i}(x)dx = \int_{\partial\Omega} u(s)n_i(s)ds$.

3.2.2 The variational formulation

Define :

1. a bilinear form $a : (u, v) \mapsto \int_\Omega \nabla u \cdot \nabla v$
2. a linear form $\ell : v \mapsto \int_\Omega f v$

Since u vanishes on the boundary, let us choose $H = H_0^1(\Omega)$ endowed with $||\cdot||_{H_0^1}$.
a, ℓ are defined on $H \subseteq H^1(\Omega)$.

We are looking for $u \in H$ such that $\forall v \in H$, $a(u, v) = \ell(v)$.

3.2.3 Continuity of a and ℓ

Our objective is to apply Lax-Milgram Theorem.
Let $(u, v) \in (H_0^1(\Omega))^2$.
Using Cauchy-Schwarz, $|a(u, v)| = \left|\int_\Omega \nabla u \cdot \nabla v\right| \leq ||\nabla u||_{L^2}||\nabla v||_{L^2} \leq ||u||_{H_0^1}||v||_{H_0^1}$.
Using Poincaré, $|\ell(v)| = \left|\int_\Omega f v\right| \leq ||f||_{L_2}|v||_{L_2} \leq C_\Omega ||f||_{L^2}||v||_{H_0^1}$.
Thus, a, ℓ are continuous on H_0^1.

3.2.4 Coercivity of a

We keep working towards our objective of applying the Lax-Milgram theorem. Let $u \in H_0^1(\Omega)$.
Then $a(u, u) = \int_0^1 ||\nabla u||_{\mathbb{R}^d}^2 = ||u||_{H_0^1}^2$.
Therefore, a is coercive on $H_0^1(\Omega)$.
Note : Here again, we did not choose the norm on H_0^1 by chance.

3.2.5 Existence and uniqueness of the weak solution

By applying the Lax-Milgram theorem, $\exists! u \in H$, $\forall v \in H$, $a(u, v) = \ell(v)$.
Furthermore, $a(u, u) = \ell(u) \implies ||u||_{H_0^1} \leq C_\Omega ||f||_{L^2}$.

3.2.6 Solving de PDE in $\mathcal{D}'$

Since $\mathcal{D}(\Omega) \subseteq H_0^1(\Omega)$, $\forall \phi \in \mathcal{D}$, $\int_\Omega \nabla u \cdot \nabla \phi = \int_\Omega f \phi$.
This is equivalent to $\forall \phi \in \mathcal{D}(\Omega)$, $\langle \nabla u, \nabla \phi \rangle = \langle f, \phi \rangle$, i.e. $\forall \phi \in \mathcal{D}(\Omega)$, $\langle -\Delta u - f, \phi \rangle = 0$.
Thus, $-\Delta u = f \in \mathcal{D}'(\Omega)$ and $u|_{\partial\Omega} = 0$ in $L^2(\Omega)$.

3.2.7 Regularity of the solution to the PDE

We have $u \in H^1(\Omega)$, which means that $u, \nabla u \in L^2(\Omega)$.
$-\Delta u = f \in L^2(\Omega)$, so $\Delta u \in L^2(\Omega)$ almost everywhere.

If Ω is a regular open set of class C^1, then the trace theorem yields $u = 0$ in $L^2(\Omega)$.

Notes :

- we cannot conclude immediately that $u \in H^2(\Omega)$ as we did in dimension 1. It is false in the general case.
- $\begin{cases} \Omega \text{ is a disk or a square} \\ \Omega \text{ is a convex polygon} \\ \Omega \text{ is the image of a convex polygon by a diffeomorphism} \end{cases} \implies u \in H^2(\Omega)$

PDE - Finite Element Method

Tom Jeandé (from John Cagnol's videos)

January 2025

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Finite Element Method (FEM) : numerical method for approximating the solution of a PDE.
It is based on

1. The variational formulation of the PDE on Ω
2. A discretization of the domain Ω
3. Replacing the PDE by its discrete counterpart
4. Solving the discretized problem by solving a system of linear equations

1 Discretization

1.1 Mesh generation

Mesh : a mesh of $\Omega \subseteq \mathbb{R}^d$ is a decomposition of Ω in subdomains (**the elements**) that do not overlap and cover Ω.
To be useful, a mesh needs to meet some quality criteria.

Dimension 1

Consider $\Omega = [a, b]$ with $a < b \in \mathbb{R}$.
For the sake of simplicity, we chose $a = 0, b = 1$, but what follows can easily be adapted.
Meshing $[0, 1]$: meshing $[0, 1]$ consists in giving

1. $J \in \mathbb{N}^*$
2. A $(J + 2)$-tuple $(x_j)_{j\in[\![0,J+1]\!]}$ such that $x_0 = 0, x_{J+1} = 1$ and $\forall j \in [\![0, J]\!], x_j < x_{j+1}$

Discretization step : the discretization step is defined by $h = \max\limits_{j\in[\![0,J]\!]} |x_{j+1} - x_j|$

If $\forall j \in [\![0, J]\!]$, $x_{j+1} - x_j$ is constant, then the mesh is said to be **uniform**, and $h = \dfrac{1}{J+1}, x_j = jh$.

Dimension 2

Let $\Omega \subseteq \mathbb{R}^2$ be a bounded set with a polygonal boundary.
Triangulation : a triangulation of Ω is given by a set $\mathcal{T}$ of J_e triangles $(K_i)_{i\in[\![1,J_e]\!]}$ such that :

1. $\forall i \in [\![1, J_e]\!]$, $K_i \subseteq \overline{\Omega}$
2. $\overline{\Omega} = \bigcup_{i\in[\![1,J_e]\!]} K_i$
3. $\forall i, j \in [\![1, J_e]\!]$, $i \neq j \Longrightarrow K_i \cap K_j$ is either empty, a vertex or an edge

The verticles of the triangles in $\mathcal{T}$ are called **nodes**.
In order to be suitable, the triangles need to meet some additional properties depending on the finite element simulation.

Diameter of a triangle : the diameter of a triangle K is given by $\mathrm{diam}(K) = \sup_{x,y\in K} |x - y|$.
We can then define the **size of the mesh** : $h = \max_{i\in[\![1,J_e]\!]} \mathrm{diam}(K_i)$.
Futhermore, we denote by :

- J_e the number of triangles
- J_v the number of nodes
- J_v^D the number of nodes on $\partial\Omega$

Dimension d

Let $\Omega \subseteq \mathbb{R}^d$ be a bounded set with a polyhedral boundary.
Triangulation : a triangulation of Ω is given by a set $\mathcal{T}$ of J_e *simplices* $(K_i)_{i\in[\![1,J_e]\!]}$ such that :

1. $\forall i \in [\![1, J_e]\!]$, $K_i \subseteq \overline{\Omega}$
2. $\overline{\Omega} = \bigcup_{i\in[\![1,J_e]\!]} K_i$
3. $\forall i, j \in [\![1, J_e]\!]$, $i \neq j \Longrightarrow K_i \cap K_j$ is either empty, or a simplex of dimension equal or smaller that $d - 1$

The verticles of the triangles in $\mathcal{T}$ are called **nodes**.

1.2 Internal approximation

Consider a variational formulation (VF) : find $u \in H$, $\forall v \in H$, $a(u, v) = \ell(v)$, where

- H is a Hilbert space
- a a continuous and coercive bilinear form on $H \times H$
- ℓ is a continuous linear form on H

Let us consider a collection of linear subspaces $H_h \subseteq H$ of finite dimension N_h.
The parameter h has no significance at this point. We will choose it smaller and smaller later.

Notes :

1. H_h endowed with the inner product of H is a Hilbert space
2. a, ℓ are respectively a continous coercive bilinear form on $H_h \times H_h$ and a continuous linear form on H_h

Consider (VF_h) : find $u_h \in H_h$, $\forall v_h \in H_h$, $a(u_h, v_h) = \ell(v_h)$.
Well-posedness of (VF_h) : by applying the Lax-Milgram theorem, we find that (VF_h) is well-posed.

Let $(\phi_i)_{i\in[\![1,N_h]\!]}$ be a basis of H_h.
Rigidity matrix : the rigidity matrix A_h is a $N_h \times N_h$ matrix such that $A_h = \big[a(\phi_i, \phi_j)\big]_{(i,j)\in[\![1,N_h]\!]^2}$
Properties of the rigidity matrix :

1. A is invertible

2. a symmetric $\Longrightarrow A$ positive definite

Denote $b_h = \left[\ell(\phi_i)\right]_{i\in[\![1,N_h]\!]}$
Characterization of the solution to (VF_h) : the solution u_h of (VF_h) is the unique solution to $A_h u_h = b_h$

Céa's Lemma : denote by $\alpha > 0$ a constant which satisfies this coercivity a.
Then $||u - u_h||_H \leq \dfrac{M}{\alpha} \inf\limits_{v_h \in H_h} ||u - v_h||_H$.

Consequences of the symmetry of a : assume that a is symmetric. Then:

1. $||\cdot||_E = \sqrt{a(\cdot,\cdot)}$ is a norm, called the **energy norm**
2. $||\cdot||_E$ is equivalent to $||\cdot||_H$, the norm of the Hilbert space : $\sqrt{\alpha}||\cdot||_H \leq ||\cdot||_E \leq M||\cdot||_H$
3. u_h is the orthogonal projection of u on H_h, and $||u - u_h||_H \leq \sqrt{\dfrac{M}{\alpha}} \inf\limits_{v_h \in H} ||u - v_h||$

The Céa Lemma provides an upper-bound to the error made when replacing u by u_h. This upper-bound depends on :

1. α, the coercivity constant of a
2. M, the continuity constant of a
3. How "far away" u is from H_h

As the dimension of H_h increases, we can expect the error to become smaller and smaller.

Convergence of the internal approximation method :
Let $(H_h)_h$ be a sequence of N_h-dimensional linear subspaces of H.
Assume :

1. $(H_h)_h$ is decreasing, i.e. $\forall k < h \Longrightarrow H_h \subseteq H_k$
2. $\lim\limits_{h\to 0} N_h = +\infty$
3. $\exists \mathcal{H}$ dense linear subspace of H, $r_h : \mathcal{H} \to H_h$ such that $\forall v \in \mathcal{H}, \lim\limits_{h\to 0} ||v - r_h(v)||_H = 0$

Then :

1. the internal approximation method converges, i.e. $\lim\limits_{h\to 0} ||u - u_h||_H = 0$
2. $||u - u_h||_H = O(h^p) \Longrightarrow$ the method is of order p

r_h is called the **interpolation operator**.

2 Implementing the FEM

2.1 Outline of the method

Outline of the method

1. Find a "good" decreasing sequence of spaces $(H_h)_h$
2. Solve the linear system in H_h to find u_h
3. $\lim\limits_{h\to 0} u_h$ is a solution to (VF)

But how can we find the "good" spaces H_h?

1. We must build the interpolation operator $r_h : \mathcal{H} \to H_h$ such that $\forall v \in \mathcal{H}, \lim_{h\to 0} ||v - r_h(v)||_H = 0$
2. We would like the rigidity matrix to be sparse (with many 0) and behave "well" for solving $A_h u_h = b_h$

The FEM consists of putting together spaces of piecewise polynomial continuous functions.
If H is separable then $\exists (e_k)_{k\in\mathbb{N}}$ Hilbertian basis of H.
Let $h = \dfrac{1}{n}$ and $H_h = \text{Span}(e_k,\, k \in [\![1, n]\!])$.
Let r_h be the orthogonal projection from H to H_h.

Finding such basis needs to be done carefully so that $A_h u_h = b_h$ can be done easily. We would like :

1. A_h to be as sparse as possible, to store less data and speed up the computation
2. $\kappa_h = ||A_h|| \cdot ||A_h^{-1}||$ to be as small as possible, and not to diverge. κ_h is called the **condition number**, and it measures the sensitivity of the solution u_h to small errors in b_h

$P_q = \{\text{Span}(X_1^{\alpha_1} \dots X_d^{\alpha_d}, \sum_{k=1}^{d} \alpha_k \leq q)\}$ is the space of polynomials of degree smaller than q in the variables $X_1, \dots X_d$

2.2 Finite elements P_1 in dimension 1

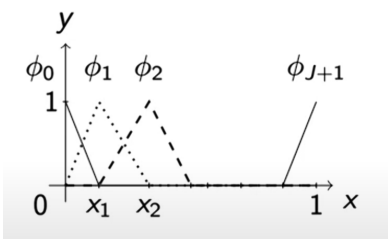
Again, we consider the interval $[0, 1]$ and a mesh $(x_j)_{j\in[\![0,J+1]\!]}$ on this interval.

Finite elements P_1 : we define $H_h = \{v \in C^0([0, 1]),\, \forall j \in [\![0, J]\!],\, v|_{[x_j,x_{j+1}]} \in P_1\}$, as well as its subspace $H_{0,h} = \{v \in H_h,\, v(0) = v(1) = 0\}$.

Properties of $H_h, H_{0,h}$:

1. H_h is a linear subspace of $H^1(0, 1)$ of dimension $J + 2$
2. $H_{0,h}$ is a linear subspace of $H_0^1(0, 1)$ of dimension J

$\forall j \in [\![0, J + 1]\!]$, ϕ_j is an affine by part function which is 0 on $[0, x_{j-1}] \cup [x_{j+1}, 1]$ and 1 at x_j.



Basis of $H_h, H_{0,h}$:

1. $(\phi_j)_{j\in[\![0,J+1]\!]}$ is a basis of H_h
2. $(\phi_j)_{j\in[\![1,J]\!]}$ is a basis of $H_{0,h}$

Solving $-u'' = f$ with homogenous Dirichlet boundary conditions

We consider :

1. a uniform mesh $(x_j)_{j\in[\![0,J+1]\!]}$ with $J \geq 1$ and $h = \dfrac{1}{J+1}$
2. the space $H_{0,h} \subseteq H_0^1(0, 1)$ with basis $(\phi_j)_{j\in[\![1,J]\!]}$
3. the variational formulation (VF_h) : find $u_h \in H_h$, $\forall v_h \in H_h$, $a(u_h, v_h) = \ell(v_h)$

Let $A_h = \left[a(\phi_i, \phi_j)\right]_{(i,j)\in[\![1,J]\!]^2} = \left[\int_0^1 \phi_i'\phi_j'\right]_{(i,j)\in[\![1,J]\!]^2} = \begin{cases} \frac{2}{h} \text{ if } i = j \\ \frac{-1}{h} \text{ if } |i-j| = 1 \\ 0 \text{ otherwise} \end{cases}$.

Let $b_h = \left[\int_0^1 f\phi_i\right]_{i\in[\![1,J]\!]}$.

Interpolation operator P_1 : linear mapping $r_h : \begin{cases} H^1(0,1) \to H_h \\ v \mapsto \sum_{j=0}^{J+1} v(x_j)\phi_j \end{cases}$

Convergence of the FEM P_1 : let $u \in H_0^1(0,1)$, $u_h \in H_{0,h}$ be solutions to $(VF), (VF_h)$ respectively. Then :

1. the FEM P^1 converges, i.e. $\lim_{h\to 0} ||u - u_h||_{H^1} = 0$
2. $\exists C > 0$, $u \in H^2(0,1) \Longrightarrow ||u - u_h||_{H^1} \leq Ch||f||_{L^2}$

The result of the computation shows that we already have a great approximation for $J = 10$.

2.3 Finite elements P_1 in dimension 2

Let $\Omega \subseteq \mathbb{R}^2$, and denote by $\partial\Omega$ its boundary.
Finite elements P_1 : we define $H_h = \{v \in C^0(\Omega), \forall j \in [\![0,J]\!], v|_{K_j} \in P_1\}$, as well as its subspace $H_{0,h} = \{v \in H_h, v|_{\partial\Omega} = 0\}$.

Properties of $H_h, H_{0,h}$:

1. H_h is a linear subspace of $H^1(\Omega)$ of dimension J_v
2. $H_{0,h}$ is a linear subspace of $H_0^1(\Omega)$ of dimension J_v^D

Define $(\phi_j)_{j\in[\![0,J_v+1]\!]}$ such that : $\begin{cases} \forall j \in [\![1,J_v]\!], \phi_j \in H_h \\ \forall i,j \in [\![1,J_v]\!], \phi_j(x_i) = \delta_{ij} \end{cases}$.

Basis of $H_h, H_{0,h}$:

1. $(\phi_j)_{j\in[\![1,J_v]\!]}$ is a basis of H_h
2. $(\phi_j)_{j\in[\![1,J_v^D]\!]}$ is a basis of $H_{0,h}$

Solving $-\Delta u = f$ on Ω with homogeneous Dirichlet boundary conditions

We consider :

1. a triangulation $\mathcal{T}$ of Ω
2. the space $H_{0,h} \subseteq H_0^1(\Omega)$ with basis $(\phi_j)_{j\in[\![1,J_v^D]\!]}$
3. the variational formulation (VF_h) : find $u_h \in H_h$, $\forall v_h \in H_h$, $a(u_h, v_h) = \ell(v_h)$

Let $A_h = \left[a(\phi_i, \phi_j)\right]_{(i,j)\in[\![1,J_v^D]\!]^2} = \left[\int_\Omega \nabla\phi_i\nabla\phi_j\right]_{(i,j)\in[\![1,J_v^D]\!]^2}$ and $b_h = \left[\ell(\phi_i)\right]_{i\in[\![1,J_v^D]\!]}$.
We can thus carry out the computations in the exact same way as in dimension 1.

Let $K = (A_1A_2A_3)$ be a non-degenerate triangle in $\mathbb{R}^2$.
$\{A_1, A_2, A_3\}$ is **P_1-unisolvent**, i.e. given $(f_1, f_2, f_3) \in \mathbb{R}^3$, there exists a unique $p \in P_1$ such that $\forall i \in \{1,2,3\}$, $p(A_i) = f_i$.

Barycentric coordinates : $\forall M = (x,y) \in \mathbb{R}^2$, $\exists!(\lambda_K^1, \lambda_K^2, \lambda_K^3) \in \mathbb{R}^3$, $\begin{cases} \sum_{i=1}^3 \lambda_K^i = 1 \\ M = \sum_{i=1}^3 \lambda_K^i A_i \end{cases}$.

The triplet $(\lambda_K^1, \lambda_K^2, \lambda_K^3)$ is called the barycentric coordinates of M. They are linear in (x,y).
It is a natural basis of P_1 on the triangle K.
The barycentric coordinates system depends on the triangle.

3 FEM software and libraries

3.1 A non-exhaustive list

Free software libraries :

- FreeFem++
- FEniCS
- SfePy (for use with Python)
- JulaFEM
- MFEM

Commercial software libraries :

- Comsol Multiphysics
- ANSYS
- ADINA
- Abaqus
- LS-DYNA (crash analysis)
- Nastran

3.2 An example with FEniCS

Useful resource : `https://cagnol.link/fenicsbook`
See video for a tutorial

PDE - Finite Difference Method

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Finite Difference Method : the core idea of the method is to :

1. Mesh the domain to obtain a grid
2. Approximate the derivatives of u using the values of u on the grid
3. Have a finite number of unknowns (the values of u on the grid) and to solve a linear system.

1 The Finite Difference Method in dimension $d = 1$

1.1 Approximating the derivative

Let $b > aa$, and f be a real-valued function defined on $[a, b]$.
Let $J \in \mathbb{N}^*$ and $(x_j)_{j \in [\![0, J+1]\!]}$ be a mesh of $[a, b]$.
Denote by $h = \dfrac{b-a}{J+1}$.

Forward difference at $x = x_j$: $f(x+h) - f(x) = f(x_{j+1}) - f(x_j)$
Backward difference at $x = x_j$: $f(x) - f(x-h) = f(x_j) - f(x_{j-1})$

Central difference at x : $f(x + h/2) - f(x - h/2)$
Note : $x \pm h/2$ are not on the grid ! We thus usually take the average of the two nearest points on the grid.

- $f(x + h/2)$ is usually replaced with $\dfrac{f(x+h) + f(x)}{2}$
- $f(x - h/2)$ is usually replaced with $\dfrac{f(x-h) + f(x)}{2}$

The **central difference at $x = x_j$** is thus defined by $\dfrac{f(x_{j+1}) - f(x_{j-1})}{2}$

Forward difference quotient : $\dfrac{f(x+h) - f(x)}{h} = \dfrac{f(x_{j+1}) - f(x_j)}{x_{j+1} - x_j}$
Backward difference quotient : $\dfrac{f(x) - f(x-h)}{h} = \dfrac{f(x_j) - f(x_{j-1})}{x_j - x_{j-1}}$
Central difference quotient : $\dfrac{f(x+h/2) - f(x-h/2)}{h} = \dfrac{f(x_{j+1}) - f(x_{j-1})}{x_{j+1} - x_{j-1}}$

Local truncation error/Discretization error : difference between the difference quotient and the derivative.
If the local truncation error is $O(h^k)$, the method is of order k.

Approximation using usual quotients :

- The forward difference quotient approximates the derivative and the method is of order 1.
- The backward difference quotient approximates the derivative and the method is of order 1.
- The central difference quotient approximates the derivative and the method is of order 2.

Stencil : the stencil of a numerical approximation method is the geometric arrangement of the nodes that relate to the point of interest.
In dimension 1, the stencil is "flat", and we may have more complex arrangemetns in dimension $d > 1$.

Notes :

- difference quotients may also be called Newton quotients, Fermat difference quotients or finite difference approximation of the derivative.
- forward finite differences are sometimes denoted by Δ, backward differences by ∇, central differences by δ. These notations won't be used in this class.

1.2 Approximating higher-order derivatives

One can obtain finite difference approximations to higher order derivatives by applying the finite difference approximation to f'.

Second-order central difference quotient : $\dfrac{f(x+h) - 2f(x) + f(x-h)}{h^2}$
Approximation of the second derivative : the second(order difference quotient approximates the second derivative. The method is of order 2.

n^{th}-order central difference quotient : $\dfrac{1}{h^n}\sum_{i=0}^{n}(-1)^i \binom{n}{i} f(x + (\frac{n}{2} - i)h)$.

n^{th}-forward difference quotient : $\dfrac{1}{h^n}\sum_{i=0}^{n}(-1)^i \binom{n}{i} f(x + (n-i)h) = \dfrac{1}{h^n}\sum_{i=0}^{n}(-1)^i \binom{n}{i} f(x_{j+n-i})$

n^{th}-backward difference quotient : $\dfrac{1}{h^n}\sum_{i=0}^{n}(-1)^i \binom{n}{i} f(x - ih) = \dfrac{1}{h^n}\sum_{i=0}^{n}(-1)^i \binom{n}{i} f(x_{j-i})$

Approximatng the nth derivative :

- the nth-order central difference quotient approximates the nth derivative. The method is of order 2.
- the nth-order forward and backward difference quotients approximate the nth derivative. The method is of order 2.

Note : for odd n, the function is not evaluated on the grid. The problem may be remediated by taking the average of the two nearest points on the grid (just like we did for order 1).

1.3 Approximating the solutions of an ODE

Consider $f \in C^0([0,1])$, $c \in C^0([0,1], \mathbb{R}^+)$ and

$$\begin{cases} -u''(x) + c(x)u(x) = f(x) \text{ in } (0,1) \\ u(0) = u(1) = 0 \end{cases}$$

Consider a uniform mesh $(x_j)_{j \in [\![0, J+1]\!]}$.
Define $c_j = c(x_j)$, $f_j = f(x_j)$.
Let the u_j be the unknowns that will approximate the $u(x_j)$.
We know that $u_0 = u_{J+1} = 0$ (exact values).

$$\forall j \in [\![1, J]\!],\ -u''(x_j) + c(x_j)u(x_j) = f(x_j)$$

$$\iff -\frac{1}{h^2}u_{j-1} + \left(\frac{2}{h^2} + c_j\right)u_j - \frac{1}{h^2}u_{j+1} = f_j$$

Denote by $A_h = \frac{1}{h^2}\begin{bmatrix} 2 & -1 & & & 0 \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ 0 & & & -1 & 2 \end{bmatrix}$, $C_h = \begin{bmatrix} c_1 & & & & 0 \\ & c_2 & & & \\ & & \ddots & & \\ & & & c_{J-1} & \\ 0 & & & & c_J \end{bmatrix}$, $F_h \begin{bmatrix} f_1 \\ \vdots \\ f_J \end{bmatrix}$, $U_h = \begin{bmatrix} u_1 \\ \vdots \\ u_J \end{bmatrix}$

Then $(A_h + C_h) \cdot U_h = F_h$.

Existence and unicity of the solution : $\forall J \geq 1$, $(A_h + C_h) \cdot U_h = F_h$ has a unique solution U_h.

The matrix A_h is called the **FDM-Matrix of the Laplacian** in dimension 1.

Non-negative vector : a vector v is non-negative if all of its components are non-negative. We note $v \geq 0$.
Non-negative matrix : a matrix M is non-negative if all of its components are non-negative. We note $M \geq 0$.

Monotone matrix : a matrix $M \in \mathcal{M}_q(\mathbb{R})$ is monotone if $\forall v \in \mathbb{R}^q$, $Mv \geq 0 \implies v \leq 0$.
Properties of monotone matrices :

1. A monotone matrix is non singular.
2. M monotone $\iff M^{-1} \geq 0$.

$A_h + C_h$ **is monotone**.
Maximum principle : if $f \geq 0$, then U_h has only non-negative components.

1.4 Convergence

U_h is not a discretization of the solution. It does *not* hold the values of u at the nodes of the grid. We only want to *approximate* $u(x_j)$ with u_j.
We discretized the problem, not the solution !

Projection onto the mesh : the projection Π_h of a function onto the mesh is defined by its pointwise evaluation on the nodes $(x_j)_j$:

$$\Pi_h : \begin{cases} C([0,1]) \to \mathbb{R}^J \\ u \mapsto [u(x_j)]_{j \in [\![1,J]\!]} \end{cases}$$

We have $F_h = \Pi_h(f)$, $C_h = \Pi_h(c)$.
However, we do not have $U_h = \Pi_h(u)$ in general. We thus want to measure the **gap** :

$$E_h = U_h - \Pi_h(u) = \begin{bmatrix} u_1 - u(x_1) \\ \vdots \\ u_J - u(x_J) \end{bmatrix}$$

Convergence : the numerical method is convergent if $\lim_{h \to 0} ||E_h||_h = 0$, with $||\cdot||_h$ a norm on $\mathbb{R}^J$.
Furthermore, if $||E_h||_h = O(h^p)$, the method is of **order** p.

Consistency : define $\mathcal{E}_h = A_h \Pi_h(u) - F_h \in \mathbb{R}^J$.
The FDM is consistent with the PDE if $\lim_{h \to 0} \max |(E_h)_j| = \lim_{h \to 0} ||\mathcal{E}_h||_\infty = 0$.
Furthermore, the consistency is of **order** p if $||\mathcal{E}_h||_\infty = O(h^p)$.
In other words, "consistency $\iff \lim_{\text{mesh size} \to 0} (PDE - FDM) = 0$" (no mathematical sense)

Stability : consider a numerical method $A_h U_h = F_h$ where U_h is the unknown, F_h the input data and A_h a non singular matrix.
Consider $|||\cdot|||$ the matrix norm induced by the norm $||\cdot||$ on $\mathbb{R}^J$.
The numerical method is stable if there exists a constant C independent of J such that $|||A_h^{-1}||| \leq C$.
Stability means that small errors in the input data F_h will not get out of hand.

Note : stability depends on the chosen norm, since C has to be independent of J.

Sufficient condition for convergence : a FDM that is both stable and consistent is convergent.

Corollary : the FDM for the Poisson's problem in dimension 1 converges in $||\cdot||_\infty$, and the method is of order 2.

2 FDM in dimension $d > 1$

2.1 Meshing

For the FEM, we often mesh using simplices (triangles in 2D).
Howver for the FDM, we rather mesh with orthotopes (rectangles in 2D). The mesh follows a pattern structure : we tesselate Ω.

Structured mesh : mesh that can be produced by replicating an elementary cell.
A mesh that is not structured is called **unstructured**.
With a structured mesh, every vertex can easily be numbered by a d-tuple, with d being the dimension of the considered space.

2.2 Approximating partial derivatives

We can generalize what we did in lower dimensions. For first order dervatives :

$$\frac{\partial f}{\partial x}(x,y) = \frac{f(x+h,y) - f(x,y)}{h} + O(h) \quad \text{(forward difference quotient, } x \text{ component)}$$

$$\frac{\partial f}{\partial y}(x,y) = \frac{f(x,y+k) - f(x,y)}{k} + O(k) \quad \text{(forward difference quotient, } y \text{ component)}$$

$$\frac{\partial f}{\partial x}(x,y) = \frac{f(x,y) - f(x-h,y)}{h} + O(h) \quad \text{(backward difference quotient, } x \text{ component)}$$

$$\frac{\partial f}{\partial y}(x,y) = \frac{f(x,y) - f(x,y-k)}{k} + O(k) \quad \text{(backward difference quotient, } y \text{ component)}$$

$$\frac{\partial f}{\partial x}(x,y) = \frac{f(x+\frac{h}{2},y) - f(x-\frac{h}{2},y)}{h} + O(h^2) \quad \text{(central difference quotient, } x \text{ component)}$$

$$\frac{\partial f}{\partial y}(x,y) = \frac{f(x,y+\frac{k}{2}) - f(x,y-\frac{k}{2})}{k} + O(k^2) \quad \text{(central difference quotient, } y \text{ component)}$$

For second order derivatives (we only write central difference quotients here) :

$$\frac{\partial^2 f}{\partial x^2}(x,y) = \frac{f(x+h,y) - 2f(x,y) + f(x-h,y)}{h^2} + O(h^2)$$

$$\frac{\partial^2 f}{\partial y^2}(x,y) = \frac{f(x,y+k) - 2f(x,y) + f(x,y-k)}{k^2} + O(k^2)$$

$$\frac{\partial^2 f}{\partial x \partial y}(x,y) = \frac{f(x+\frac{h}{2},y+\frac{k}{2}) + f(x-\frac{h}{2},y-\frac{k}{2})}{hk} - \frac{f(x+\frac{h}{2},y-\frac{k}{2}) + f(x-\frac{h}{2},y+\frac{k}{2})}{hk} + O(h^2,k^2).$$

2.3 Approximating the solution of a PDE

Consider a well-posed PDE, for instance :

$$f \in C^0(\Omega), \begin{cases} -\Delta u = f \text{ on } \Omega \\ u|_{\partial\Omega} = 0 \end{cases}$$

Let us discretize Δu : $\Delta u \approx \dfrac{1}{h^2}\left(u_{j,i-1} + u_{j-1,i} - 4u_{j,i} + u_{j+1,i} + u_{j,i+1}\right)$.
We can express this approximation using a single index by considering $(i,j) \mapsto n = J(i-1) + j$:

$$-\Delta u \approx \frac{1}{h^2}\left(-u_{j+J(i-2)} - u_{j-1+J(i-1)} + 4u_{j+J(i-1)} - u_{j+1+J(i-1)} - u_{j+Ji}\right)$$

$$= \frac{1}{h^2}\left(-u_{n-J} - u_{n-1} + 4u_n - u_{n+1} - u_{n+J}\right)$$

with u_n replaced by zero for all out-of-range n. Therefore,

$$\frac{1}{h^2}\left(-u_{n-J} - u_{n-1} + 4u_n - u_{n+1} - u_{n+J}\right) = f_{j+J(i-1)} = f_n$$

Define A_h such that :

$$A_h = \frac{1}{h^2}\begin{bmatrix} T & -I & 0 & \cdots & 0 \\ -I & T & -I & \cdots & \vdots \\ 0 & -I & T & \ddots & 0 \\ \vdots & \vdots & \ddots & \ddots & -I \\ 0 & 0 & 0 & -I & T \end{bmatrix}$$

with

$$T = \begin{bmatrix} 4 & -1 & 0 & \cdots & 0 \\ -1 & 4 & -1 & \cdots & \vdots \\ 0 & -1 & 4 & \ddots & 0 \\ \vdots & \vdots & \ddots & \ddots & -1 \\ 0 & 0 & 0 & -1 & 4 \end{bmatrix}, \quad I = \begin{bmatrix} 1 & 0 & \cdots & \cdots & 0 \\ 0 & 1 & \cdots & \cdots & \vdots \\ \vdots & \vdots & \ddots & \ddots & 0 \\ 0 & 0 & \ddots & 1 & 0 \\ 0 & 0 & \cdots & 0 & 1 \end{bmatrix}$$

(*Refer to the video for the code*)

3 FEM-FDM comparison

FDM	**FEM**
Consistency error *Measures how well the FDM represents the original solution*	Céa's Lemma *Provides error bounds, ensuring that the FEM solution is as close as possible to the best approximation in the finite element space*
Consistency *Ensures that the discrete scheme converges to the exact differential equation as the grid is refined.*	Interpolation *Process of approximating a function using finite element basis functions.*
L^∞ stability *Control over the maximum error in the numerical solution.*	L^2 stability *Refers to the stability of the numerical solution in terms of the L^2 norm, which measures the overall energy of the error.*

Table 1: Comparison between the two methods

While the FDM focuses on local discretization errors and stability in maximum norms, FEM uses more global approximation concepts and typically measures stability in energy norms.

PDE7 - Numerical linear algebra

Tom Jeandé

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1 Norms on a matrix

1.1 Definition

Let $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$.
$E = \mathcal{M}_{q,p}(\mathbb{K})$ is a $\mathbb{K}$-linear space.

Norm : mapping $N : E \to \mathbb{R}_+$ which satisfies :

1. **Separation** : $N(x) = 0 \iff x = 0$
2. **Homogeneity** : $\forall \lambda \in \mathbb{K},\ \forall x \in E,\ N(\lambda x) = |\lambda| N(x)$
3. **Triangle inequality** : $\forall x, y \in E,\ N(x+y) \leqslant N(x) + N(y)$

For a matrix, the norm is usually denoted by $|||\cdot|||$.

Subordinance : $|||\cdot|||$ is subordinated if

$$\forall A \in \mathcal{M}_{q,p}(\mathbb{K}), \|Ax\| \leqslant |||A||| \, \|x\|$$

Submultiplicativity : $|||\cdot|||$ is submultiplicative if

$$\forall A, B \in \mathcal{M}_{q,q}(\mathbb{K}),\ |||AB||| \leqslant |||A||| \, |||B|||$$

Induced norm : $\sup\limits_{x \neq 0} \dfrac{\|Ax\|}{\|x\|}$ is a norm on E, called the induced norm. It depends on the norm defined on $\mathbb{K}$. This norm indicates the maximum gain of A.

Properties of the induced norm :

1. The induced norm is subordinated
2. If $p = q$, the induced norm is submultiplicative

Norm infinity : the norm infinity of a vector is given by $\|x\|_\infty = \max\limits_{1\leqslant i\leqslant q} |x_i|$.

The induced matrix norm $\|\cdot\|_\infty$ is defined by $|||\cdot|||_\infty : A \mapsto \max\limits_{1\leqslant i\leqslant q} \sum\limits_{j=1}^{q} |a_{ij}|$, which is to the maximum absolute row sum of the matrix.

Norm 1 : the norm 1 of a vector is given by $\|x\|_1 = \sum\limits_{i=1}^{q} |x_i|$.

The induced matrix norm $|||\cdot|||_1$ is defined by $|||A|||_1 = \max\limits_{1\leqslant j\leqslant q} \sum\limits_{i=1}^{q} |a_{ij}|$, which is to the maximum absolute column sum of the matrix.

Conjugate transpose : the conjugate transpose of $A \in \mathcal{M}_{q,p}(\mathbb{C})$ is obtained by transposing A, then taking the conjugate of each entry. It is denoted by A^*.
Singular value : square root of the eigenvalues of A^*A, denoted by $\sigma_i(A)$.

Norm 2 : the norm 2 of a vector is given by $\|x\|_2 = \sqrt{\sum\limits_{i=1}^{q} x_i^2}$.

The induced matrix norm $|||\cdot|||_2$ is defined by $|||A|||_2 = \sigma_{max}(A)$, which is the largest singular value of the matrix A.

Froebenius inner product : E can be endowed with the inner product

$$\langle A, B\rangle = \sum_{i=1}^{p}\sum_{j=1}^{q} \bar{A}_{ij}B_{ij}$$

This inner product is called the Froebinius inner product.

Froebinius norm : the Froebinius norm is defined by $|||A|||_F = \sqrt{\langle A, A\rangle}$, with $\langle\cdot,\cdot\rangle$ being the Froebinius inner product.
Properties of the Froebinius norm :

1. If $p = q$, the Froebinius norm is submultiplicative
2. Assume that $p = q$, and let $|||\cdot|||_2$ be the norm induced by $\|\cdot\|_2$ on $\mathbb{R}^q$.
   Then $|||\cdot|||_2 \leqslant |||\cdot|||_F$
   This implies that $\forall A \in \mathcal{M}_q(\mathbb{R}),\ \forall x \in \mathbb{R}^q,\ \|Ax\|_2 \leqslant |||A|||_F\, \|x\|$ (useful since $|||\cdot|||_F$ is easy to compute)

1.2 Stability of the FDM-Matrix of the Laplacian in dimension 1

About the matrix associated to Poisson's equation in dimension 1 : let A_h be the matrix associated to Poisson's equation in dimension 1. Then $\forall J \geqslant 1,\ |||A_h^{-1}|||_\infty \leqslant \frac{1}{8}$ (see proof in the video)

1.3 Conditioning of a matrix

Condition number : let $\|\cdot\|$ be a norm on $\mathbb{K}^q$ and $|||\cdot|||$ be the induced norm on $\mathcal{M}_q(\mathbb{K})$, and let $A \in GL_q(\mathbb{K})$.
The condition number of A relative to $|||\cdot|||$ is defined by $\kappa(A) = |||A|||\ |||A^{-1}|||$.
The condition number is always greater or equal to one : $\kappa(A) \geqslant 1$

Bounding the error using the condition number : consider $A \in GL_q(\mathbb{K})$, and consider $b_1, b_2 \in \mathbb{R}^q$.
Let $x_1 = A^{-1}b_1,\ x_2 = A^{-1}b_2$. Then :

$$\frac{\|x_2 - x_1\|}{\|x_1\|} \leqslant \kappa(A)\frac{\|b_2 - b_1\|}{\|b_1\|}$$

where $\kappa(A)$ is the condition number relative to the matrix norm induced by $\|\cdot\|$.
When solving a linear system of equations, we thus want the condition number to be as close to 1 as possible.

Preconditioner : the preconditioner P of a matrix A is a matrix such that $\kappa(PA) < \kappa(A)$.
For a non-singular matrix, the best preconditioner is $P = A^{-1}$ (but if A^{-1} is known, the problem was solved in the first place...). Finding a good P means finding preconditioner that is easy to compute and which yields

good results.

Condition number relative to $\|\cdot\|_2$ for positive-definite matrix : let $A \in \mathcal{M}_q(\mathbb{R})$ be a positive-definite matrix with eigenvalues $0 < \lambda_1 < \cdots < \lambda_q$.
The condition number κ_2 relative to the matrix norm induced by $\|\cdot\|_2$ is given by $\kappa_2(A) = \dfrac{\lambda_q}{\lambda_1}$

2 Spectral properties

2.1 The Gershgörin circle theorem

Any matrix $A \in \mathcal{M}_q(\mathbb{C})$ has q eigenvalues in $\mathbb{C}$.
Spectrum : set of the eigenvalues of A in $\mathbb{C}$, denoted by $\sigma(A)$.

Unitary matrix : $U \in \mathcal{M}_q(\mathbb{C})$ is unitary if $U^{-1} = U^*$. The set of unitary matrices is denoted by $U_q(\mathbb{C})$.

Schur decomposition : let $A \in \mathcal{M}_q(\mathbb{C})$. Then there exists :

1. an upper triangular matrix T
2. a unitary matrix U

such that $A = UTU^{-1}$

Let $A \in \mathcal{M}_q(\mathbb{C})$, we denote by A_{ij} the components of A.

Gerschgörin discs : for $i \in [\![1, q]\!]$, let $R_i = \sum_{j \neq i} A_{ij}$.
The closed disk $D_i = B(A_{ii}, R_i) \subseteq \mathbb{C}$ is called a Gerschgörin disc.

Gerschgörin theorem : $\sigma(A) \subseteq \bigcup_{k=1}^{q} D_k$

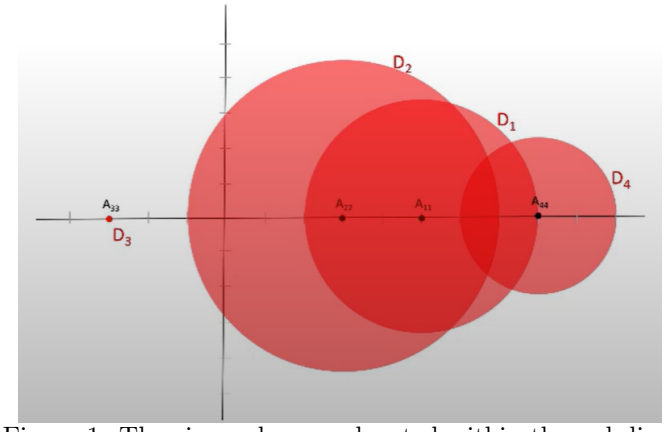

Figure 1: The eigenvalues are located within the red discs

When looking for a preconditioner P for matrix A, the eigenvalues of PA should all be close to 1. The Gerschgörin circle theorem yields that every eigenvalue of PA lies within a know area.

2.2 The Power Iteration Method

Spectral radius : let $A \in \mathcal{M}_q(\mathbb{C})$, the spectral radius of A is the non-negative number

$$\rho(A) = \max\{|\lambda|,\ \lambda \in \sigma(A)\}$$

Property of the spectral radius : let $\|\cdot\|$ be an induced norm on $\mathcal{M}_q(\mathbb{C})$, then $\forall A \in \mathcal{M}_q(\mathbb{C}),\ \rho(A) \leqslant \|A\|$.
Note : $\forall A \in \mathcal{M}_q(\mathbb{C}),\ |||A|||_2 = \sqrt{\rho(A^*A)}$. If A is symmetric positive definite, $|||A|||_2 = \rho(A)$.

The Power Method : let $A \in \mathcal{M}_q(\mathbb{C})$, and $\|\cdot\|$ be a norm on $\mathbb{K}^q$.
Let $\lambda_1, \dots, \lambda_q$ be the eigenvalues of A, with $|\lambda_1| > |\lambda_2| \geqslant \dots \geqslant |\lambda_q|$.
Let e_1 be an eigenvector associated to λ_1, and $F = Im(A - \lambda_1 I_q)$.
Define $(x_n)_n$ by

$$x_{n+1} = \frac{Ax_n}{\|Ax_n\|}$$

and $x_0 = \mu e_1 + f$, with $f \in F$ and $\mu \neq 0$.
Then

$$\lim_{n\to\infty} \|Ax_n\| = \rho(A)$$

x_0 has to be chosen to have a non-zero component along e_1. We apply A to x_n, then normalize it. Intuitively, the dominant eigenvalue will "pull" towards the direction of e_1. In practice, x_0 can be chosen randomly since the probability that we pick a vector orthogonal to e_1 is negligible (and the system would be very unstable due to numerical precision).

3 Iteration Matrices

Linear recurrence relation : let $M \in \mathcal{M}_q(\mathbb{K})$, and $b, x_0 \in \mathbb{K}^q$.
The sequence $(x_n)_{n\in\mathbb{N}}$ defined by $x_{n+1} = Mx_n + b$ is called a linear recurrence relation.
The matrix M is called the **iteration matrix**.

Convergent numerical method : consider a linear recurrence relation with the iteration matrix M and a non-empty set of possible initial conditions $C \subseteq \mathbb{K}^q$.

$$\begin{cases} x_0 \in C \\ \forall n \geqslant 0,\ x_{n+1} = Mx_n + b \end{cases}$$

The sequence is a convergent numerical method if for all $b \in \mathbb{K}^q$, $(x_n)_{n\in\mathbb{N}}$ converges.

Convergence rate : consider a convergent numerical method $(x_n)_{n\in\mathbb{N}}$.
Denote by $x = \lim x_n$ and $e_n = x_n - x$ for all $n \geqslant 0$.
The convergence rate of $(x_n)_{n\in\mathbb{N}}$ is a measure of the decline of $(e_n)_{n\in\mathbb{N}}$ towards 0.

Convergence of numerical methods : the 4 following statements are equivalent :

1. $\forall x \in \mathbb{K}^q$, $(x_n)_{n\in\mathbb{N}}$ defined by $\begin{cases} x_0 = x \\ \forall n \geqslant 0,\ x_{n+1} = Mx_n \end{cases}$ converges towards 0.
2. $\lim\limits_{n\to\infty} M^n = 0$
3. $\rho(M) < 1$
4. There exists an induced norm $|||\cdot|||$ such that $|||M||| < 1$

4 Solving linear systems

4.1 Direct methods

***LU* decomposition** : $A \in GL_q(\mathbb{K})$ has a LU decomposition if there exists :

1. a lower triangular matrix L with 1 on the diagonal
2. an upper triangular matrix U

such that $A = LU$.
Pros :

- Very useful if we have to solve $Ax = b$ for several b
- Sparse matrices stay sparse in the process

Cons : the complexity is q^3.

Note : we solve $Ax = b$, we do not invert A.

Existence of the LU decomposition : there exists a LU decomposition if the Gauss elimination works out. In that case, it is unique.

Cholesky decomposition theorem : let $A \in GL_q(\mathbb{R})$ be symmetric positive definite.
There exists a lower triangular matrix B with positive diagonal terms such that $B^*B = A$.

QR decomposition : every matrix $A \in GL_q(\mathbb{R})$ can be uniquely decomposed as $A = QR$, where $Q \in O_q(\mathbb{R})$ and R are upper triangular matrices with positive terms on the diagonal.

4.2 Iterative methods

4.2.1 Solving a numerical system using an iterative method

This process consists of two main steps :

1. Decomposing A in $A = M - N$, where M is a non-singular matrix.
2. Considering the numerical method $\begin{cases} x_0 \in \mathbb{K}^q \\ \forall n \geqslant 0,\ Mx_{n+1} = Nx_n + b \end{cases}$

Note : at each iteration, one needs to solve a system. M thus needs to be *chosen* so that the system is simple.

Convergence of the numerical method : $\rho(M^{-1}N) < 1 \Longrightarrow$ the numerical method converges.

A few possibilities for choosing M and N

Jacobi : $M = diag(A)$ and $N = -A + M$.

Gauss-Seidel : consider

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1q} \\ a_{21} & a_{22} & \cdots & a_{2q} \\ \vdots & \vdots & \ddots & \vdots \\ a_{q1} & a_{q2} & \cdots & a_{qq} \end{bmatrix}$$

This method consists in taking

$$M = \begin{bmatrix} a_{11} & 0 & \cdots & 0 \\ a_{21} & a_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ a_{q1} & a_{q2} & \cdots & a_{qq} \end{bmatrix},\ N = -\begin{bmatrix} 0 & a_{12} & \cdots & a_{1q} \\ 0 & 0 & \cdots & a_{2q} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix}$$

In other words, A is the lower triangular part of M (diagonal included) and N is the opposite of the upper triangular part of M (diagonal excluded).

4.2.2 Conjugate Gradient

Conjugate Gradient : this method consists in turning our problem into an optimization problem.

$$Ax = b \Longleftrightarrow \|Ax - b\| = 0$$

The goal is thus to minimize $F : x \mapsto \|Ax - b\|$.
See more about this method in the optimization course next year !

PDE8 - Parabolic PDEs

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The aim of this chapter is to tackle parabolic PDEs. Indeed, the numerical methods that we've covered in previous chapter only seem to apply to elliptic PDEs.

1 Wellposedness

1.1 Some functional spaces

Let $\Omega \subset \mathbb{R}^d$ be an bounded open set.
Let $f : (t,x) \mapsto f(t,x)$ an $u^0 : x \mapsto u^0(x)$ be given functions.
Consider the heat equation :

$$\begin{cases} \partial_t u(t,x) - \alpha \Delta u(t,x) = f(t,x), \ \forall (t,x) \in \mathbb{R}_+^* \times \Omega \\ u(\cdot, x) = 0, \ \forall x \in \partial\Omega \\ u(0,\cdot) = u^0 \end{cases}$$

We can "separate" the variables : u is seen as

$$u : \begin{cases} \mathbb{R}_+^* \to X \\ t \mapsto u(t) \end{cases}$$

with X a space a functions from Ω to $\mathbb{R}_+^*$.

Let $T > 0$, and X be a Hilbert space.
Denote by $\langle \cdot, \cdot \rangle_X$ its inner product, and $\| \cdot \|$ its norm.
Space $L^2((0,T),X)$: space of functions v such that $t \mapsto \|v(t)\|_X^2$ is integrable.

Properties of $L^2((0,T),X)$:

1. $\begin{cases} L^2((0,T),X) \to \mathbb{R} \\ v \mapsto \displaystyle\int_0^T \|v(t)\|^2 dt \end{cases}$ is a norm on $L^2((0,T),X)$.
2. $v \in L^2((0,T),X) \iff \|v\|_{L^2((0,T),X)}$ is finite.

3. $L^2((0,T),X)$ is a Hilbert space for the inner product $\langle\cdot,\cdot\rangle_{L^2((0,T),X)} : (f,g) \mapsto \int_0^T \langle f(t), g(t)\rangle_X dt$

Let $k \in \mathbb{N}^*$.
Space $C^k([0,T],X)$: space of functions v such that $t \mapsto \|v(t)\|_X^2$ is of class C^k.

C^k is a Banach space : C^k is a Banach space for the norm

$$\boxed{f \mapsto \|f\|_{C^k} = \sum_{m=0}^{k} \sup_{t\in[0,T]} \left\| \frac{d^m f}{dt^m}(t) \right\|_X}$$

1.2 Solving the heat equation

Solution of the heat equation :
Let $d \in \mathbb{N}^*$, and $\Omega \subset \mathbb{R}^d$ be a bounded open set.
Let $T > 0$, $F \in L^2(\Omega) \times L^2((0,T), L^2(\Omega))$, and $(u^0, f) \in F$.
The heat equation has a unique solution in $E = L^2((0,T), H_0^1(\Omega)) \cap C^0([0,T], L^2(\Omega)$.
Futhermore, there exists a constant $C > 0$ such that

$$\forall t, \|u(t)\|_{L^2(\Omega)}^2 + \int_0^t \|u(s)\|_{H_0^1(\Omega)}^2 ds \leqslant C \left(\|u(0)\|_{L^2(\Omega)}^2 + \int_0^t \|f(s)\|_{L^2(\Omega)}^2 ds \right)$$

The data (u_0, F) is given by:

- The initial condition u_0, quantified by its L^2-norm
- The right hand side f, quantified by the sum of its L^2-norm over time

Therefore, at time t, we can bound $u \in E$, which means that we have:

- The solution $u(t)$ in norm L^2
- The "sum" of the H_0^1 norm of u over time

In other words, the norm (in E) of the solution is *completely* controlled by the norm (in F) of the data.

Energy space: space $E = L^2((0,T), H_0^1(\Omega)) \cap C^0([0,T], L^2(\Omega))$.
The norm of the solution in E will be called the **energy**.

Notes:

1. if $f = 0$, the energy of the solution decreases, and the equation is said to be **dissipative**.
   Example: the heat equation is dissipative.
2. The "energy" defined here does not always correspond to the true physical energy of the system.

2 Qualitative properties

2.1 Regularity

Consider the following problem:

$$(\mathcal{P}) : \begin{cases} \partial_t u - \alpha \Delta u = f \text{ in } \mathbb{R}_+^* \times \Omega \\ u(\cdot, x) = 0, \ \forall x \in \partial\Omega \\ u(0, \cdot) = u^0 \end{cases}$$

Question: Can we increase the regularity on u?

We already showed that $(u^0, f) \in F = L^2(\Omega) \times L^2((0,T), L^2(\Omega))$ gives a unique solution $u \in E = L^2((0,T), H_0^1(\Omega)) \cap C^0([0,T], L^2(\Omega))$.
Let's see if increasing the regularity on f and u^0 allows us to increase the regularity on u.

Regularity with no source term: let $\Omega \subset \mathbb{R}^d$ be a bounded regular open set.
Assume that $f = 0$.
Let $u_0 \in L^2(\Omega)$, and u be the solution to the problem $(\mathcal{P})$.

Then $u \in C^\infty(\mathbb{R}_+^* \times \overline{\Omega})$.
Note that we have a **regularizing effect**: the solution u is smoother than the data u_0.

Regularity when u_0 is more regular: let $\Omega \subset \mathbb{R}_+^*$ be a bounded regular open set.
Let $(u_0, f) \in H_0^1(\Omega) \times L^2((0,T), L^2(\Omega))$.
Let $u \in E$ be the solution to the problem $(\mathcal{P})$.
Then $u \in L^2((0,T), H^2(\Omega)) \cap C^0([0,T], H_0^1(\Omega))$ and $\partial_t u \in L^2((0,T), L^2(\Omega))$.

2.2 Asymptotic behavior

We are now interested in the behavior of $u(t)$ for large t.
The behavior will depend of f:

Asymptotic behavior for f constant: let Ω be a bounded regular open set.
Let $(u^0, f) \in F$, and assume that **f does not depend on t**.
Let $u \in E$ be the solution to $(\mathcal{P})$.
Then u converges to the solution $\overline{u}$ to the steady state problem:

$$\begin{cases} -\alpha \Delta \overline{u} = f \text{ in } \Omega \\ \overline{u}(x) = 0, \ \forall x \in \partial\Omega \end{cases}$$

Asymptotic behavior for $f = 0$: let Ω be a bounded regular open set.
Let $u_0 \in L^2(\Omega)$ and assume that $f = 0$.
Let $u \in E$ be the solution to $(\mathcal{P})$ under these hypotheses.
Then $\|u(t)\|_{L^2(\Omega)} \xrightarrow[t \to +\infty]{} 0$.

2.3 Maximum principle

Maximum principle: let $\Omega \subset \mathbb{R}^d$ be a bounded regular open set.
Let $(u^0, f) \in F$, and assume that $f \geqslant 0$ almost everywhere in $(0,T) \times \Omega$ and $u_0 \geqslant 0$ almost everywhere in Ω.
Let $u \in E$ be the solution to $(\mathcal{P})$ under these hypotheses.
Then $u \geqslant 0$ almost everywhere in $[0,T] \times \Omega$.
Physical interpretation: if the initial temperature is positive and we keep heating the room, then the temperature will remain positive.

Propagation at infinite speed: let $\Omega \subset \mathbb{R}^d$ be a bounded regular open set.
Let $(u_0, f) \in F$ and assume that $f \geqslant 0$ almost everywhere in $(0,T) \times \Omega$ and $u^0 \geqslant 0$ almost everywhere in Ω.
Furthermore, assume that u_0 is not identically zero.
Let $u \in E$ be the solution to $(\mathcal{P})$.
Then $\forall \varepsilon > 0, \ \forall x \in \Omega, u(\varepsilon, x) > 0$.
Meaning of the above proposition: if $f = 0$ and $u^0 \geqslant 0$ with $u^0 \neq 0$, then $\forall t \in (0,T], \ \forall x \in \Omega, \ u(t,x) > 0$.
In other words, we have an infinite speed of propagation: impossible physically
$\Longrightarrow$ a model is a simplification of the world.

Note; the question "Is the model true?" is never a good question : a model is always a simplification.
The good question to ask is "Is the model useful?" Does it capture the information I need for the application that I'm considering.

3 Numerical approximation

3.1 Semi-discretization using the FDM

Let $\Omega = (0,1), \ u^0 \in L^2(0,1), \ f \in C^0(\mathbb{R}^+ \times [0,1])$.
Let $J \in \mathbb{N}^*, \ h_x = \dfrac{1}{J+1}$ and consider the uniform mesh $(x_j)_{j \in [\![1,J]\!]}$ on $(0,1)$.
Consider the heat equation:

$$(H) : \begin{cases} \partial_t u - \alpha \partial_{xx}^2 u = f \text{ in } \mathbb{R}_+^* \times (0,1) \\ u(\cdot, 0) = u(\cdot, 1) = 0 \\ u(0, \cdot) = u^0 \end{cases}$$

Let us write the **space semi-discretization**:

$$(H_D) : \begin{cases} \forall t \geqslant 0,\ \forall j \in [\![1, J]\!],\ v_j'(t) - \alpha \dfrac{v_{j+1}(t) - 2v_j(t) + v_{j-1}(t)}{h_x^2} = f(t, x_j) \\ \forall t \geqslant,\ v_0(t) = v_{J+1}(t) = 0 \\ \forall j \in [\![1, J]\!],\ v_j(0) = u^0(x_j) \end{cases}$$

In Chapter VI, we introduced the FDM matrix for the Laplace-operator.
Let A_{h_x} be this matrix in dimension $d = 1$, and consider:

$$V : t \mapsto \begin{bmatrix} v_1(t) \\ \vdots \\ v_J(t) \end{bmatrix},\ U^0 = \begin{bmatrix} u^0(x_1) \\ \vdots \\ u^0(x_J) \end{bmatrix},\ \mathcal{F} : t \mapsto \begin{bmatrix} f(t, x_1) \\ \vdots \\ f(t, x_J) \end{bmatrix}$$

The problem can then simply be written as:

$$\boxed{\begin{cases} V'(t) = -\alpha A_{h_x} V(t) + \mathcal{F}(t) \\ V(0) = U^0 \end{cases}}$$

keeping in mind that $v_0(t) = v_J(t) = 0$.

This system is a first-order linear IVP of dimension J.

Solving a fisrt-order linear IVP: for $A \in \mathcal{M}_J(\mathbb{R}),\ t \mapsto b(t),\ U^0 \in \mathbb{R}^d$, consider the system

$$\begin{cases} V'(t) = -\alpha A V(t) + \mathcal{F}(t) \\ V(0) = U^0 \end{cases}$$

This system has a unique solution on $\mathbb{R}$, given by

$$\boxed{V : t \mapsto \exp(-\alpha t A) U^0 + \int_0^t \exp(-\alpha(t - s)A)\mathcal{F}(s) ds}$$

The Lyapunov Corollary can be adapted to $\mathcal{F}(t)$: convergence of

$$\begin{cases} V'(t) = -\alpha A_{h_x} V(t) + \mathcal{F}(t) \\ V(0) = U^0 \end{cases}$$

is linked to the spectrum of $-\alpha t A_{h_x}$.
Spectrum of A_{h_x}: the spectrum of A_{h_x} is

$$\boxed{\sigma(A_{h_x}) = \left\{ \frac{4}{h_x^2} \sin^2\left(\frac{\pi j}{2(J+1)}\right),\ j \in [\![1, J]\!] \right\}}$$

About the spectral radius of A_{h_x}: $\rho(A_{h_x}) = \max \sigma(A_{h_x}) \leqslant \dfrac{4}{h_x^2}$

Smallest eigenvalue of A_{h_x}: $\lambda_1 = \min \sigma(A_{h_x}) = \dfrac{4}{h_x^2} \sin^2\left(\dfrac{\pi h_x}{2}\right)$

Note: $\lambda_1 = \pi + O(h_x)$

Asymptotic behavior: let $\mathcal{F}$ be independent of t, and $\overline{V}$ be the solution to $-\alpha A_{h_x} \overline{V} + \mathcal{F} = 0$.
Then

$$\boxed{\forall t \geqslant 0, \|V(t) - \overline{V}\|_{\mathbb{R}^J, 2} \leqslant \exp(-\alpha \lambda_1 t) \|V(0) - \overline{V}\|_{\mathbb{R}^J, 2}}$$

where

- $\|\cdot\|_{\mathbb{R}^J, 2}$ is the Euclidian norm on $\mathbb{R}^J$
- $\lambda_1 = \min \sigma(A_{h_x})$

Corollary: $\lim_{t\to\infty} V(t) = \overline{V}$

Until now, we have proven that $V(t)$ converges to $\overline{V}$ as $t \to \infty$, but not that $V(t)$ tends to $u(t)$ as Δx tends to 0 or that $\overline{V}$ tends to $\overline{u}$ as Δx tends to 0.

For $u \in C^0([0,T] \times [0,1])$, denote by $\Pi_{h_x} u(t) = \begin{bmatrix} u(t,x_1) \\ \vdots \\ u(t,x_J) \end{bmatrix}$

Convergence of the Finite Differences: the solution to the semi-discretized problem converges to the continuous problem. The order of convergence is 2.

Let $T > 0$, V be the the solution to the semi-discrete problem and u be the solution to the continuous problem. Then there exists $C > 0$ depending only on T and u^0 such that

$$\boxed{\sup_{t\in[0,T]} \|V(t) - \Pi_{h_x} u(t)\|_{\mathbb{R}^J,2} \leqslant C h_x^2}$$

3.2 Semi-discretization using the FEM

Let $\Omega \subset \mathbb{R}^d$ be a polyhedron.
Consider the variational formulation :

$$\text{Find } u \in L^2((0,T), H_0^1(\Omega)),\ \forall v \in H_0^1(\Omega),\ \frac{d}{dt}\int_\Omega u(t,x)v(x)dx + \alpha \int_\Omega \nabla u(t,x)\cdot\nabla v(x)dx = \int_\Omega f(t,x)v(x)dx$$

Let $\mathcal{T}_h$ be a mesh with:

- n_S interior verticles $(S_j)_{j\in[\![1,n_S]\!]}$
- n_T triangles $(T_i)_{i\in[\![1,n_T]\!]}$

We use the P_1 method with a basis $(\phi_j)_{j\in[\![1,n_S]\!]}$.
We look for a solution $u_h : t \mapsto \sum_{j=1}^{n_S} u_j(t)\phi_j$ to

$$\begin{cases} \forall i \in [\![1,n_S]\!],\ \sum_{j=1}^{n_S} u_j'(t)\int_\Omega \phi_j\phi_i + \alpha \sum_{j=1}^{n_S} u_j(t)\int_\Omega \nabla\phi_j\cdot\nabla\phi_i = \int_\Omega f\phi_i, \\ \forall j \in [\![1,n_S]\!],\quad u_j(0) = u^0(S_j). \end{cases}$$

Rigidity/Stiffness matrix: $n_S \times n_S$ matrix such that

$$\boxed{(\mathcal{A}_h)_{ij} = \int_\Omega \nabla\phi_j\cdot\nabla\phi_i}$$

Mass matrix: $n_S \times n_S$ matrix such that

$$\boxed{(\mathcal{M}_h)_{ij} = \int_\Omega \phi_j\phi_i}$$

Denote by

$$U_h = \begin{bmatrix} u_1(t) \\ \vdots \\ u_{n_S}(t) \end{bmatrix},\ \mathcal{F}_h = \begin{bmatrix} \int_\Omega f\phi_1 \\ \vdots \\ \int_\Omega f\phi_{n_S} \end{bmatrix}$$

The equation of the variational formulation can be rewritten:

$$\forall i \in [\![1,n_S]\!],\ \sum_{j=1}^{n_S}(U_h')_{ij}(\mathcal{M}_h)_{ij} + \alpha\sum_{j=1}^{n_S}(U_h)_j(\mathcal{A}_h)_{ij} = (\mathcal{F}_h)_{ij}$$

$$\iff \forall i \in [\![1,n_S]\!],\ (\mathcal{M}_h U_h')_i + \alpha(\mathcal{A}_h U_h)_i = (F_h)_i$$

$$\iff \boxed{\mathcal{M}_h U_h' + \alpha\mathcal{A}_h U_h = \mathcal{F}_h}$$

This last equation, with $U_h(0)$ given, is an Initial Value Problem.
Just like before, we have global solutions.
However, just like before, we still need a time discretization.

3.3 Time-space discretization

Let us apply the Explicit Euler method to the discretization:

$$\begin{cases} V' = -\alpha \mathcal{A}_h V + b_h \\ V(0) = V^0 \end{cases}$$

Let $T > 0$. Consider a uniform discretization of $[0, T]$: let $N \geqslant 1,\ h_t \geqslant \dfrac{T}{N}$ and $t^n = nh_t$ for $n \in [\![0, N]\!]$.
For all $n \in [\![0, N]\!]$, denote by $b_h^n = b_h(t^n)$. We have:

$$\begin{cases} V^0 = V(0) \\ \forall n \in [\![0, N]\!],\ V^{n+1} = V^n - h_t \alpha \mathcal{A}_h V^n + h_t b_h^n \end{cases}$$

This sequence is an arithmetico-geometric sequence.
$\forall n \in [\![0, N]\!]$, we have

$$V^n = (I_J - h_t \alpha \mathcal{A}_h)^n V^0 + h_t \sum_{k=0}^{n-1} (I_J - h_t \alpha \mathcal{A}_h)^{n-1-k} b_h^k$$

Stability: let $J \geqslant 1$.
A fully discretized method (in space and time)

$$W^{n+1} = BW^n + d^n$$

is stable in the space norm $\|\cdot\|$ on $\mathbb{R}^J$ if there exists $C > 0$ such that

$$\boxed{\forall N \geqslant 1, \forall n \in [\![0, N]\!],\ \forall J \geqslant 1,\ \|B^n\|_{\mathcal{M}_J(\mathbb{R})} \leqslant C}$$

C may depend on T, but is independent of J and N.
Sufficient condition for stability: a sufficient condition for stability in norm $\|\cdot\|_{\mathbb{R}^J,2}$ of the method

$$V^{n+1} = (I_J - h_t \alpha A_{h_x}) V^n + h_t F^n$$

is

$$\boxed{\rho(I_J - h_t \alpha A_{h_x}) \leqslant 1}$$

We now want a sufficient condition to have $\rho(I_J - h_t \alpha A_{h_x}) \leqslant 1$.
We have $\rho(I_J - h_t \alpha A_{h_x}) \leqslant -1 + h_t \alpha \rho(A_{h_x})$ (or all the eigenvalues are in $(0, 1)$.
So

$$\rho(I_J - h_t \alpha A_{h_x}) \leqslant -1 + 4\alpha \frac{h_t}{h_x^2}$$

A sufficient condition is

$$4\alpha \frac{h_t}{h_x^2} \leqslant 2$$

This condition can be simplified into the following condition:
CFL Condition: the inequality

$$\boxed{\alpha \frac{h_t}{h_x^2} \leqslant \frac{1}{2}}$$

is called the CFL condition of the explicit method, centered in space forward in time, for the heat equation, in $\|\cdot\|_{\mathbb{R}^J,2}$.

Note: each PDE may have a different CFL.

The CFL condition describes how the time step h_t should be chosen with respect to the spatial discretization step h_x in order to avoid any problem.

Consistency: the consistency error for the explicit method, centered in space, is defined $\forall n \in [\![0, N-1]\!]$ by:

$$\mathcal{E}^n = \frac{\Pi_{h_x} u(t^{n+1}) - \Pi_{h_x} u(t^n)}{h_t} + \alpha A_{h_x} \Pi_{h_x} u(t^n) - F_{h_x}^n$$

Using Taylor-Lagrange:

$$\max \|\mathcal{E}_n\|_\infty = O(h_t) + O(h_x^2)$$

Consistency and Stability imply Convergence:
Let u be the exact solution to the heat equation.
Let $h_x,\ h_t$ be chosen to meet the CFL condition $\alpha h_t \leqslant h_x^2/2$.
Then $(V^n)_n$ converges towards u, and the orders of convergence are 2 in space and 1 in time.
In $\|\cdot\|_2$, $\forall N \geqslant 0,\ \forall J \geqslant 1$ such that $\alpha h_t \leqslant h_x^2/2$,

$$\forall n \in [\![0, N]\!],\ \|V^n - \Pi_{h_x} u(t^n)\|_2 = O(h_x^2) = O(h_t)$$

Furthermore, under the CFL condition, the maximum principle and the energy dissipativity can be established.

Note: for the *explicit* method, the CFL requirement imposes $h_t \leqslant \dfrac{h_x^2}{2}$, which is a tough requirement on h_t.
Instead of the explicit method, we could have used an implicit method, which tend to be more stable but requires solving an equation at each step.